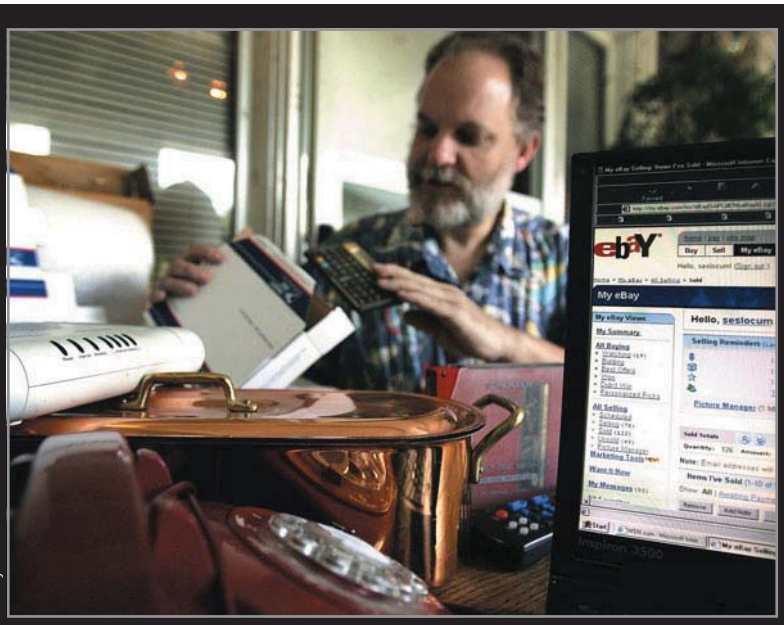


Don Ryan/AP Photos

3

DEVELOPING A PROCESS STRATEGY

At any given time eBay has approximately 113 million listings worldwide, and yet its workforce consists of just 15,000 employees. The explanation? Customers do most of the work in eBay's buying and selling processes. Here a customer prepares items for shipping from sales on his Ebay account.

eBay

Most manufacturers do not have to contend with customers waltzing around their shop floors, showing up intermittently and unannounced. Such customer contact can introduce considerable variability, disrupting carefully designed production processes. Costs and quality can be adversely affected. While customer contact is an issue even with manufacturers, (each process does have at least one customer), extensive customer contact and involvement are business as usual for many processes of service providers. Customers at restaurants or rental car agencies are directly involved in performing the processes. The area where the sales person interacts with the customer *is* the shop floor.

How much should customers be involved in a process, so as to provide timely delivery and consistent quality, and at sustainable cost? Various ways are available—some accommodate customer-introduced variability and some reduce it. eBay illustrates one way to accommodate variability. As an online auction house, eBay has high volume and request variability. Its customers do not want service at the same time or at times necessarily convenient to the company. They have request variability, seeking to buy and sell an endless number of items. They also have variability in customer capability, some with considerable Internet experience and some needing more handholding. Such variability would greatly complicate workforce scheduling if eBay required its employees to conduct all of its processes. It connects hundreds of millions of people around

the world every day. It has a global presence in 39 markets, with revenue of $9.2 billion in more than 50,000 categories—and only with 15,500 employees. This relatively small workforce is possible in the face of customer-induced variability because its customers perform virtually all of the selling and buying processes through the eBay Web site. When the customer is responsible for much of the work, the right labor is provided at the right moment.

Source: Frances X. Frei, "Breaking the Trade-Off between Efficiency and Service," *Harvard Business Review* (November 2006), pp. 93–101; **http://en.wikipedia.org/wiki/Ebay** (March 19, 2011).

process strategy
The pattern of decisions made in managing processes so that they will achieve their competitive priorities.

LEARNING GOALS *After reading this chapter, you should be able to:*

1. Explain why processes exist everywhere in all organizations.
2. Discuss the four major process decisions.
3. Position a process on the customer-contact matrix or product-process matrix.
4. Configure operations into layouts.
5. Define customer involvement, resource flexibility, capital intensity, and economies of scope.
6. Discuss how process decisions should fit together.
7. Define process reengineering and process improvement.

Competing with Operations
Using Operations to Compete
Managing Effective Projects
Designing and Managing Processes
Developing a Process Strategy
Analyzing Processes
Managing Quality
Planning Capacity
Managing Process Constraints
Designing Lean Systems
Designing and Managing Supply Chains
Managing Inventories
Designing Effective Supply Chains
Locating Facilities
Integrating the Supply Chain
Managing Sustainable Supply Chains
Forecasting Demand
Planning and Scheduling Operations
Planning Sufficient Resources

Process decisions, such as the amount of customer involvement allowed at eBay, are strategic in nature: As we saw in Chapter 1, they should further a company's long-term competitive goals. In making process decisions, managers focus on controlling such competitive priorities as quality, flexibility, time, and cost. Process management is an ongoing activity, with the same principles applying to both first-time and redesign choices.

In this chapter, we focus on **process strategy**, which specifies the pattern of decisions made in managing processes so that the processes will achieve their competitive priorities. Process strategy guides a variety of process decisions, and in turn is guided by operations strategy and the organization's ability to obtain the resources necessary to support them. We begin by defining four basic process decisions: (1) process structure (including layout), (2) customer involvement, (3) resource flexibility, and (4) capital intensity. We discuss these decisions for both service and manufacturing processes. We pay particular attention to ways in which these decisions fit together, depending on factors such as competitive priorities, customer contact, and volume. We conclude with two basic change strategies for analyzing and modifying processes: (1) process reengineering and (2) process improvement.

Three principles concerning process strategy are particularly important:

1. The key to successful process decisions is to make choices that fit the situation and that make sense together. They should not work at cross-purposes, with one process optimized at the expense of other processes. A more effective process is one that matches key process characteristics and has a close *strategic fit.*
2. Although this section of the text focuses on individual processes, they are the building blocks that eventually create the firm's whole supply chain. The cumulative effect on customer satisfaction and competitive advantage is huge.
3. Whether processes in the supply chain are performed internally or by outside suppliers and customers, management must pay particular attention to the interfaces between processes. Dealing with these interfaces underscores the need for cross-functional coordination.

Process Strategy across the Organization

As we explained in Chapter 1, processes are everywhere and are the basic unit of work. Consider the following two major points: (1) supply chains have processes and (2) processes are found throughout the whole organization, and not just in operations.

Supply Chains Have Processes

Parts 2 and 3 of this book about operations management are titled Managing Processes and Managing Supply Chains. If you infer that both Parts are essential aspects of operations management, you are correct. If you also infer that only Part 2 deals with processes, you would be wrong. The correct

conclusion is that Part 3 is about managing **supply chain processes**, which are business processes that have external customers or suppliers. Table 3.1 illustrates some common supply chain processes.

supply chain processes
Business processes that have external customers or suppliers.

TABLE 3.1 | SUPPLY CHAIN PROCESS EXAMPLES

Process	Description	Process	Description
Outsourcing	Exploring available suppliers for the best options to perform processes in terms of price, quality, delivery time, environmental issues	**Customer Service**	Providing information to answer questions or resolve problems using automated information services as well as voice-to-voice contact with customers
Warehousing	Receiving shipments from suppliers, verifying quality, placing in inventory, and reporting receipt for inventory records	**Logistics**	Selecting transportation mode (train, ship, truck, airplane, or pipeline) scheduling both inbound and outbound shipments, and providing intermediate inventory storage
Sourcing	Selecting, certifying and evaluating suppliers and managing supplier contracts	**Cross-docking**	Packing of products of incoming shipments so they can be easily sorted more economically at intermediate warehouses for outgoing shipments to their final destination

These supply chain processes should be documented and analyzed for improvement, examined for quality improvement and control, and assessed in terms of capacity and bottlenecks. These topics are not repeated in Part 3, which instead examines the broader issues in managing supply chains, but still are essential to managing supply chain processes. Supply chain processes will be only as good as the processes within the organization that have only internal suppliers and customers. Each process in the chain, from suppliers to customers, must be designed to achieve its competitive priorities and add value to the work performed.

Processes Are Not Just in Operations

Processes are found in accounting, finance, human resources, management information systems, and marketing. Organizational structure throughout the many diverse industries varies, but for the most part, all organizations perform similar business processes. Table 3.2 lists a sample of them that are outside the operations area. All of these processes must be managed. What you learn in Part 2 applies to all of these business functions.

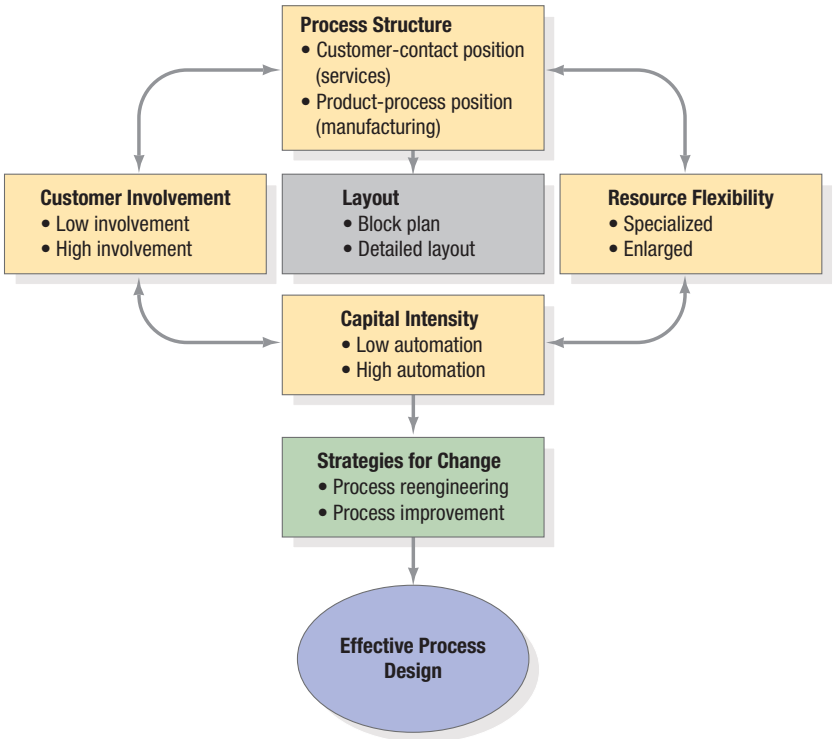
TABLE 3.2 | ILLUSTRATIVE BUSINESS PROCESSES OUTSIDE OF OPERATIONS

Activity based costing	Employee-development	Payroll
Asset management	Employee-recruiting	Records management
Billing budget	Employee-training	Research and development
Complaint handling	Engineering	Sales
Credit management	Environment	Help desks
Customer-satisfaction	External communications	Disaster recovery
Employee-benefits	Finance	Waste management
Employee-compensation	Security management	Warranty

Managers of these processes must make sure that they are adding as much customer value as possible. They must understand that many processes cut across organizational lines, regardless of whether the firm is organized along functional, product, regional, or process lines.

Process Strategy Decisions

A process involves the use of an organization's resources to provide something of value. No service can be provided and no product can be made without a process, and no process can exist without at least one service or product. One recurring question in managing processes



▲ **FIGURE 3.1**
Major Decisions for Effective Processes

is deciding *how* to provide services or make products. Many different choices are available in selecting human resources, equipment, outsourced services, materials, work flows, and methods that transform inputs into outputs. Another choice is which processes are to be done in-house, and which processes are to be outsourced—that is, done outside the firm and purchased as materials and services. This decision helps to define the supply chain, and is covered more fully in subsequent chapters. Supplement A, "Decision Making Models," also introduces make-or-buy break-even.

Process decisions directly affect the process itself and indirectly the services and the products that it provides. Whether dealing with processes for offices, service providers, or manufacturers, operations managers must consider four common process decisions. Figure 3.1 shows that they are all important steps toward an effective process design. These four decisions are best understood at the process or subprocess level, rather than at the firm level.

- **Process structure** determines the process type relative to the kinds of resources needed, how resources are partitioned between them, and their key characteristics. A **layout**, which is the physical arrangement of operations created from the various processes, puts these decisions into tangible form.
- **Customer involvement** reflects the ways in which customers become part of the process and the extent of their participation.
- **Resource flexibility** is the ease with which employees and equipment can handle a wide variety of products, output levels, duties, and functions.
- **Capital intensity** is the mix of equipment and human skills in a process. The greater the relative cost of equipment, the greater is the capital intensity.

process structure
The process type relative to the kinds of resources needed, how resources are partitioned between them, and their key characteristics.

layout
The physical arrangement of operations created by the various processes.

customer involvement
The ways in which customers become part of the process and the extent of their participation.

resource flexibility
The ease with which employees and equipment can handle a wide variety of products, output levels, duties, and functions.

capital intensity
The mix of equipment and human skills in a process.

The concepts that we develop around these four decisions establish a framework within which we can address the appropriate process design in every situation. There is no "how to" element here in this chapter. That comes in subsequent chapters that show how to actually design the processes or modify existing ones. Instead, we establish the patterns of choices that create a good fit between the four decisions. For example, if you walk through a manufacturing facility where materials flow smoothly from one work station to the next (which we will define later to be a *line* process), you would be tempted to conclude that all processes should be line processes. They seem so efficient and organized. However, if volumes are low and the products made are customized, converting to a line process would be a big mistake. When volumes are low and products are customized, resources must be more flexible to handle a variety of products. The result is a more disorganized appearance with jobs crisscrossing in many different directions depending on the product being made. Despite appearances, this process is the best choice.

Process Structure in Services

One of the first decisions a manager makes in designing a well-functioning process is to choose a process type that best achieves the competitive priorities for that process. Strategies for designing processes can be quite different, depending on whether a service is being provided or a product is being manufactured. We begin with service processes, given their huge implication for workforce resources in industrialized countries.

Nature of Service Processes: Customer Contact

A process strategy that gets customers in and out of a fast-food restaurant quickly would not be the right process strategy for a five-star restaurant, where customers seek a leisurely dining experience. To gain insights, we must start at the process level and recognize key contextual variables associated with the process. A good process strategy for a service process depends first and foremost

on the type and amount of customer contact. **Customer contact** is the extent to which the customer is present, is actively involved, and receives personal attention during the service process. Face-to-face interaction, sometimes called a *moment of truth* or *service encounter*, brings the customer and service providers together. At that time, customer attitudes about the quality of the service provided are shaped. Table 3.3 shows several dimensions of customer contact. Many levels are possible on each of the five dimensions. The nested-process concept applies to customer contact, because some parts of a process can have low contact and other parts of a process can have high contact.

TABLE 3.3 DIMENSIONS OF CUSTOMER CONTACT IN SERVICE PROCESSES

Dimension	High Contact	Low Contact
Physical presence	Present	Absent
What is processed	People	Possessions or information
Contact intensity	Active, visible	Passive, out of sight
Personal attention	Personal	Impersonal
Method of delivery	Face-to-face	Regular mail or e-mail

Customer-Contact Matrix

The customer-contact matrix, shown in Figure 3.2, brings together three elements: (1) the degree of customer contact, (2) customization, and (3) process characteristics. The matrix is the starting point for evaluating and improving a process.

customer contact
The extent to which the customer is present, is actively involved, and receives personal attention during the service process.

Customer Contact and Customization The horizontal dimension of the matrix represents the service provided to the customer in terms of customer contact and competitive priorities. A key competitive priority is how much customization is needed. Positions on the left side of the matrix represent high customer contact and highly customized services. The customer is more likely to be present and active. The process is more likely to be visible to the customer, who receives more personal attention. The right side of the matrix represents low customer contact, passive involvement, less personalized attention, and a process out of the customer's sight.

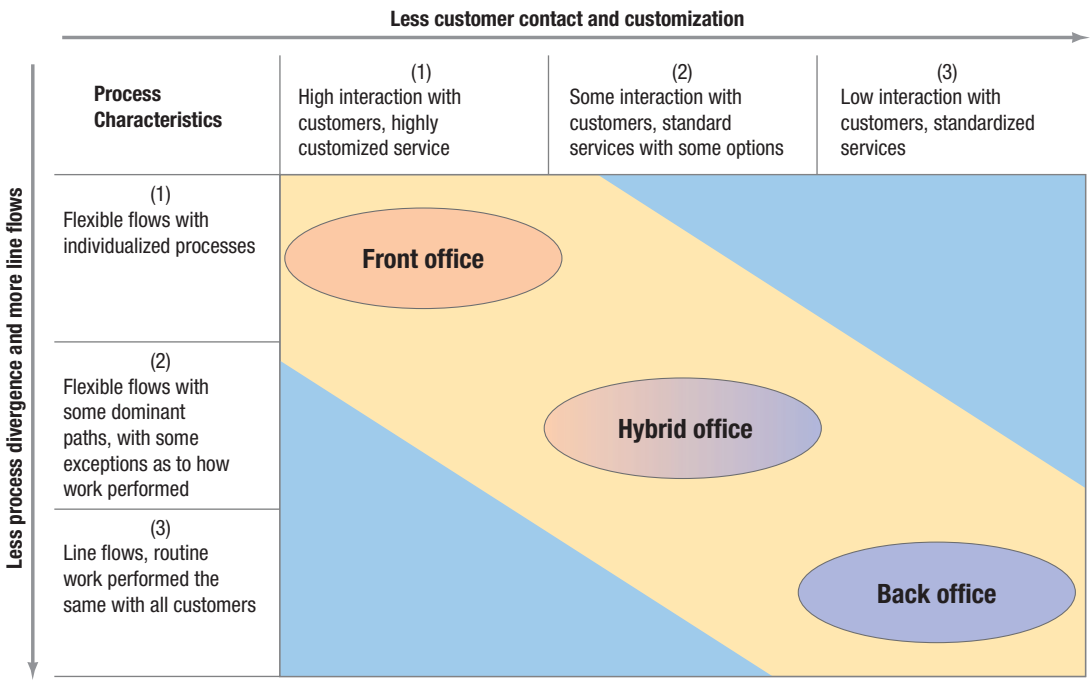
Process Divergence and Flow The vertical dimension of the customer-contact matrix deals with two characteristics of the process itself: (1) process divergence and (2) flow. Each process can be analyzed on these two dimensions.

process divergence
The extent to which the process is highly customized with considerable latitude as to how its tasks are performed.

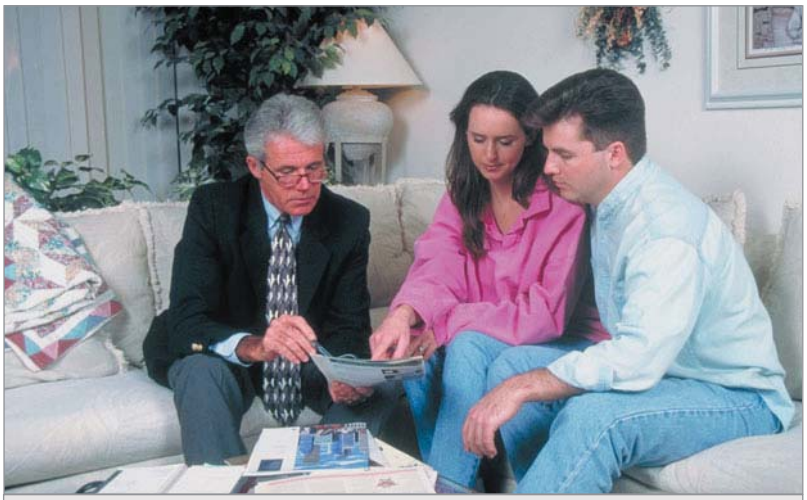
flexible flow
The customers, materials, or information move in diverse ways, with the path of one customer or job often crisscrossing the path that the next one takes.

Process divergence is the extent to which the process is highly customized with considerable latitude as to how its tasks are performed. If the process changes with each customer, virtually every performance of the service is unique. Examples of highly divergent service processes where many steps in them change with each customer are found in consulting, law, and architecture. A service with low divergence, on the other hand, is repetitive and standardized. The work is performed exactly the same with all customers, and tends to be less complex. Certain hotel services and telephone services are highly standardized to assure uniformity.

Closely related to divergence is how the customer, object, or information being processed flows through the service facility. Work progresses through the sequence of steps in a process, which could range from highly diverse to linear. When divergence is considerable, the work flow tends to be more flexible. A **flexible flow** means that the customers, materials, or information move in diverse ways,



◀ **FIGURE 3.2**
Customer-Contact Matrix for Service Processes



Myrleen Ferguson Cate/PhotoEdit

A financial consultant discusses options with a young couple at their home. This process scores high on customer contact, because the customers are present, take an active part in creating the service, receive personal attention, and have face-to-face contact.

with the path of one customer or job often crisscrossing the path that the next one takes. Each one can follow a carefully preplanned path, even though the first impression is one of disorganized, jumbled flows. Such an appearance goes naturally with high process divergence. A **line flow** means that the customers, materials, or information move linearly from one operation to the next, according to a fixed sequence. When diversity is low and the process standardized, line flows are a natural consequence.

Service Process Structuring

Figure 3.2 shows several desirable positions in the matrix that effectively connect the service product with the process. The manager has three process structures, which form a continuum, to choose from: (1) front office, (2) hybrid office, and (3) back office. It is unlikely that a process can be a top performer if a process lies too far from one of these diagonal positions, occupying instead one of the extreme positions represented by the light blue triangles in the matrix (refer to Figure 3.2). Such positions represent too much of a disconnect between the service provided and process characteristics.

line flow
The customers, materials, or information move linearly from one operation to the next, according to a fixed sequence.

front office
A process with high customer contact where the service provider interacts directly with the internal or external customer.

hybrid office
A process with moderate levels of customer contact and standard services with some options available.

back office
A process with low customer contact and little service customization.

Front Office A **front-office** process has high customer contact where the service provider interacts directly with the internal or external customer. Because of the customization of the service and variety of service options, many of the steps in it have considerable divergence. Work flows are flexible, and they vary from one customer to the next. The high-contact service process tends to be adapted or tailored to each customer.

Hybrid Office A hybrid office tends to be in the middle of the five dimensions in Table 3.3, or perhaps high on some contact measures and low on others. A **hybrid-office** process has moderate levels of customer contact and standard services, with some options available from which the customer chooses. The work flow progresses from one workstation to the next, with some dominant paths apparent.

Back Office A **back-office** process has low customer contact and little service customization. The work is standardized and routine, with line flows from one service provider to the next until the service is completed. Preparing the monthly client fund balance reports in the financial services industry is a good example. It has low customer contact, low divergence, and a line flow.

Process Structure in Manufacturing

Many processes at a manufacturing firm are actually services to internal or external customers, and so the previous discussion on services applies to them. Similarly, manufacturing processes can be found in service firms. Clarity comes when viewing work at the process level, rather than the organizational level. Here we focus instead on the manufacturing processes. Because of the differences between service and manufacturing processes, we need a different view on process structure.



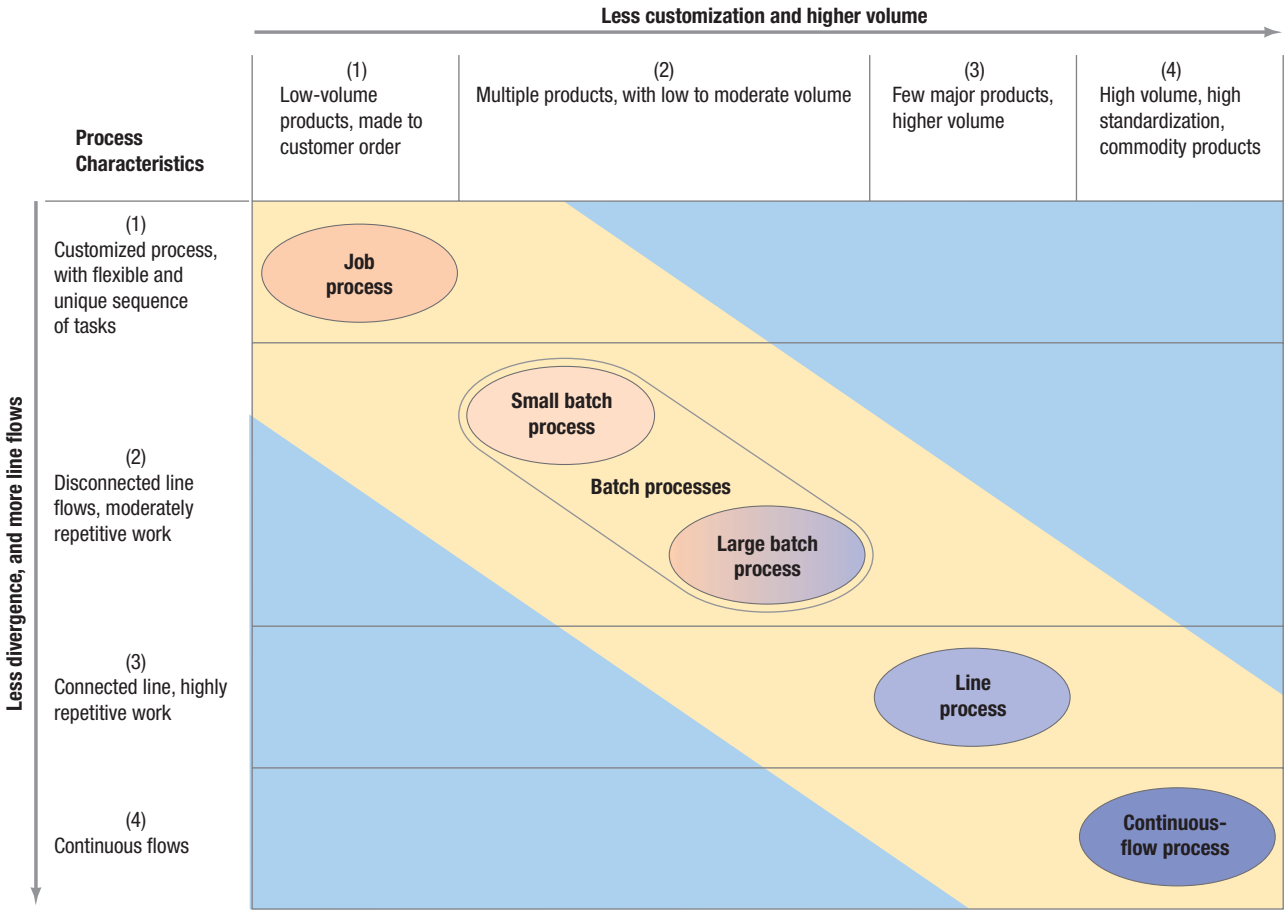
Bill Varie/Alamy

An employee discusses work with her supervisor. Each employee in this series of work stations are in a back office, because they have low customer contact and little service customization.

Product-Process Matrix

The product-process matrix, shown in Figure 3.3, brings together three elements: (1) volume, (2) product customization, and (3) process characteristics. It synchronizes the product to be manufactured with the manufacturing process itself.

A good strategy for a manufacturing process depends first and foremost on volume. Customer contact, a primary feature of the customer-contact matrix for services, normally is not a consideration for manufacturing processes (although it *is* a factor for the many service processes throughout manufacturing firms). For many manufacturing processes, high product customization means lower volumes for many of the steps in the process. The vertical dimension of the product-process matrix



▲ **FIGURE 3.3**
Product-Process Matrix for Manufacturing Processes

deals with the same two characteristics in the customer-contact matrix: process divergence and flow. Each manufacturing process should be analyzed on these two dimensions, just as was done for a service process.

Manufacturing Process Structuring

Figure 3.3 shows several desirable positions (often called *process choices*) in the product-process matrix that effectively connect the manufactured product with the process. **Process choice** is the way of structuring the process by organizing resources around the process or organizing them around the products. Organizing around the process means, for example, that all milling machines are grouped together and process all products or parts needing that kind of transformation. Organizing around the product means bringing together all the different human resources and equipment needed for a specific product and dedicating them to producing just that product. The manager has four process choices, which form a continuum, to choose from: (1) job process, (2) batch process, (3) line process, and (4) continuous-flow process. As with the customer-contact matrix, it is unlikely that a manufacturing process can be a top performer if its position is too far from the diagonal. The fundamental message in Figure 3.3 is that the best choice for a manufacturing process depends on the volume and degree of customization required of the process. The process choice might apply to an entire manufacturing process or just one subprocess nested within it.

process choice
A way of structuring the process by organizing resources around the process or organizing them around the products.

Job Process A **job process** creates the flexibility needed to produce a wide variety of products in significant quantities, with considerable divergence in the steps performed. Customization is high and volume for any one product is low. The workforce and equipment are flexible to handle considerable task divergence. Companies choosing job processes often bid for work. Typically, they make products to order and do not produce them ahead of time. Each new order is handled as a single unit—as a job. Examples are machining a metal casting for a customized order or producing customized cabinets.

job process
A process with the flexibility needed to produce a wide variety of products in significant quantities, with considerable divergence in the steps performed.

With a job process, all equipment and workers capable of certain types of work are located together. Because customization is high and most jobs have a different sequence of steps, this process choice creates flexible flows through the operations rather than a line flow.



AP Photo/David Zalubowski

A batch of apple fritters roll off one of the pastry lines at the King Soopers's Bakery in Denver to be packaged for transportation. The pastry line is a batch process, and a different kind of pastry will be made next.

Batch Process The batch process is by far the most common process choice found in practice, leading to terms such as *small batch* or *large batch* to further distinguish one process choice from another. A **batch process** differs from the job process with respect to volume, variety, and quantity. The primary difference is that volumes are higher because the same or similar products or parts going into them are produced repeatedly. Some of the components going into the final product may be processed in advance. Production lots are handled in larger quantities (or *batches*) than they are with job processes. A batch of one product (or component part going into it or perhaps other products) is processed, and then production is switched to the next one. Eventually, the first product is produced again. A batch process has average or moderate volumes, but process divergence is still too great to warrant dedicating a separate process for each product. The process flow is flexible, but more dominant paths emerge than at a job process, and some segments of the process have a line flow. Examples of a batch process are making standard components that feed an assembly line or some processes that manufacture capital equipment.

Line Process A **line process** lies between the batch and continuous processes on the continuum; volumes are high and products are standardized, which allows resources to be organized around particular products. Divergence is minimal in the process or line flows, and little inventory is held between the processing steps. Each step performs the same process over and over, with little variability in the products manufactured. Production and material handling equipment is specialized. Products created by a line process include the assembly of computers, automobiles, appliances, and toys.

Standard products are produced in advance of their need and held in inventory so that they are ready when a customer places an order. Product variety is possible by careful control of the addition of standard options to the main product.

Continuous Flow Process A **continuous flow process** is the extreme end of high-volume standardized production, with rigid line flows. Process divergence is negligible. Its name derives from the way materials move through the process. Usually, one primary material (such as a liquid, a gas, or a powder) moves without stopping through the process. A continuous-flow process differs from a line process in one important respect: Materials (be they undifferentiated or discrete) flow through the process without stopping until the whole batch is finished. The time span can be several shifts or even several months. Examples of a continuous flow process are petroleum refining; chemical processes; and processes making steel, soft drinks, and food (such as Borden's huge pasta-making plant).

batch process

A process that differs from the job process with respect to volume, variety, and quantity.

line process

A process that lies between the batch and continuous processes on the continuum; volumes are high and products are standardized, which allows resources to be organized around particular products.

continuous flow process

The extreme end of high-volume standardized production and rigid line flows, with production not starting and stopping for long time intervals.

Production and Inventory Strategies

Strategies for manufacturing processes differ from those in services, not only because of low customer contact and involvement, but also because of the ability to use inventories.[1] Make-to-order, assemble-to-order, and make-to-stock strategies are three approaches to inventory that should be coordinated with process choice.

Make-to-Order Strategy Manufacturers that make products to customer specifications in low volumes tend to use the **make-to-order strategy**, coupling it with job or small batch processes. It is a more complex process than assembling a final product from standard components. This strategy provides a high degree of customization and typically uses job or small batch processes. The processes have high divergence. Specialized medical equipment, castings, and expensive homes are suited to the make-to-order strategy.

make-to-order strategy

A strategy used by manufacturers that make products to customer specifications in low volumes.

Assemble-to-Order Strategy The **assemble-to-order strategy** is an approach to producing a wide variety of products from relatively few subassemblies and components after the customer orders are received. Typical competitive priorities are variety and fast delivery times. The assemble-to-order strategy often involves a line process for assembly and a batch process for fabrication. Because they are devoted to manufacturing standardized components and subassemblies in high volumes, the fabrication processes focus on creating appropriate amounts of component inventories for the assembly processes. Once the specific order from the customer is received, the assembly processes create the product from standardized components and subassemblies produced by the fabrication processes.

assemble-to-order strategy

A strategy for producing a wide variety of products from relatively few subassemblies and components after the customer orders are received.

Stocking finished products would be economically prohibitive because the numerous possible options make forecasting relatively inaccurate. Thus, the principle of **postponement** is applied, whereby the final activities in the provision of a product are delayed until the orders are

postponement

The strategy of delaying final activities in the provision of a product until the orders are received.

[1]Service firms also hold inventories, but only as purchased material. Manufacturing firms have the additional flexibility of holding inventories as subassemblies or finished products.

received. The assemble-to-order strategy is also linked to **mass customization**, where highly divergent processes generate a wide variety of customized products at reasonably low costs. Both postponement and mass customization are covered more fully in Chapter 10, "Designing Effective Supply Chains."

Make-to-Stock Strategy Manufacturing firms that hold items in stock for immediate delivery, thereby minimizing customer delivery times, use a **make-to-stock strategy**. This strategy is feasible for standardized products with high volumes and reasonably accurate forecasts. It is the inventory strategy of choice for line or continuous-flow processes. Examples of products produced with a make-to-stock strategy include garden tools, electronic components, soft drinks, and chemicals.

Combining a line process with the make-to-stock strategy is sometimes called **mass production**. It is what the popular press commonly envisions as the classical manufacturing process, because the environment is stable and predictable, with workers repeating narrowly defined tasks with low divergence.



James Hardy/Altopress/Newscom

A Chinese manufacturing firm using the make-to-stock strategy has considerable inventory stacked on pallets with rows of shelving racks in the background.

Layout

Selecting process structures for the various processes housed in a facility is a strategic decision, but must be followed by a more tactical decision—creating a layout. A *layout* is the physical arrangement of operations (or departments) created from the various processes and puts them in tangible form. For organizational purposes, processes tend to be clustered together into operations or departments. An *operation* is a group of human and capital resources performing all or part of one or more processes. For example, an operation could be several customer service representatives in a customer reception area; a group of machines and workers producing cell phones; or a marketing department. Regardless of how processes are grouped together organizationally, many of them cut across departmental boundaries. The flows across departmental lines could be informational, services, or products. Process structures that create more flows across departmental lines, as with job or batch processes, are the most challenging layout problems.

Here we demonstrate an approach to layout design that positions those departments close together that have strong interactions between them. It involves three basic steps, whether the design is for a new layout or for revising an existing layout: (1) gather information, (2) develop a block plan, and (3) design a detailed layout. We illustrate these steps with the Office of Budget Management (OBM), which is a major division in a large state government.

mass customization

The strategy that uses highly divergent processes to generate a wide variety of customized products at reasonably low costs.

make-to-stock strategy

A strategy that involves holding items in stock for immediate delivery, thereby minimizing customer delivery times.

mass production

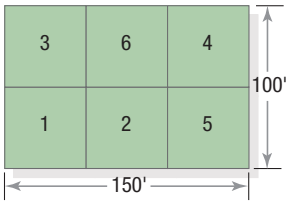
A term sometimes used in the popular press for a line process that uses the make-to-stock strategy.

Gather Information

OBM consists of 120 employees assigned to six different departments. Workloads have expanded to the extent that 30 new employees must be hired and somehow housed in the space allocated to OBM. The goal is to improve communication among people who must interact with each other effectively, creating a good work environment.

Three types of information are needed to begin designing the revised layout for OBM: (1) space requirements by center, (2) available space, and (3) closeness factors. OBM has grouped its processes into six different departments: (1) administration, (2) social services, (3) institutions, (4) accounting, (5) education, and (6) internal audit. The exact space requirements of each department, in square feet, are as follows:

Department	Area Needed (ft^2)
1. Administration	3,500
2. Social services	2,600
3. Institutions	2,400
4. Accounting	1,600
5. Education	1,500
6. Internal audit	3,400
	Total 15,000



▲ **FIGURE 3.4**
Current Block Plan for the Office of Budget Management

Management must tie space requirements to capacity and staffing plans; calculate the specific equipment and space needs for each center; and allow circulation space, such as aisles and the like. At OBM, a way must be found to include all 150 employees in its assigned area. Consulting with the managers and employees involved can help avoid excessive resistance to change and make the transition smoother.

A **block plan** allocates space and indicates placement of each operation. To describe a new facility layout, the plan need only provide the facility's dimensions and space allocations. When an existing facility layout is being modified, the current block plan is also needed. OBM's available space is 150 feet by 100 feet, or 15,000 square feet. The designer could begin the design by dividing the total amount of space into six equal blocks (2,500 square feet each). The equal-space approximation shown in Figure 3.4 is sufficient until the detailed layout stage, when larger departments (such as administration) are assigned more space than smaller departments.

block plan
A plan that allocates space and indicates placement of each operation.

The layout designer must also know which operations need to be located close to one another. The table below shows OBM's **closeness matrix**, which gives a measure of the relative importance of each pair of operations being located close together. The metric used depends on the type of processes involved and the organizational setting. It can be a qualitative judgment on a scale from 0 to 10 that the manager uses to account for multiple performance criteria, as in the OBM's case. Only the right-hand portion of the matrix is used. The closeness factors are indicators of the need for proximity based on an analysis of information flows and the need for face-to-face meetings. They give clues as to which departments should be located close together. For example, the most important interaction is between the administration and internal audit departments for OBM, with a score of 10. This closeness factor is given in the first row and last column. Thus, the designer should locate departments 1 and 6 close together, which is not the arrangement in the current layout. Entries in both the columns and rows result in five factor scores for each department.

closeness matrix
A table that gives a measure of the relative importance of each pair of operations being located close together.

CLOSENESS FACTORS

Department	1	2	3	4	5	6
1. Administration	—	3	6	5	6	10
2. Social services		—	8	1	1	
3. Institutions			—	3	9	
4. Accounting				—	2	
5. Education					—	1
6. Internal audit						—

At a manufacturing plant, the closeness factor could be the number of trips (or some other measure of materials movement) between each pair of operations per day. This information can be gleaned by conducting a statistical sampling, polling supervisors and materials handlers, or using the routings and ordering frequencies for typical items made at the plant.

Finally, the information gathered for OBM includes performance criteria that depend not on the relative location of department pairs, but the *absolute* location of a single department. OBM has two such criteria.

1. Education (department 5) should remain where it is because it is next to the office library.
2. Administration (department 1) should remain where it is because that location has the largest conference room, which administration uses often. Relocating the conference room would be costly.

Develop a Block Plan

Having gathered the needed information, the next step is to develop a block plan that best satisfies performance criteria and area requirements. The most elementary way to do so is by trial and error. Because success depends on the designer's ability to spot patterns in the data, this approach does not guarantee the selection of the best or even a nearly best solution. When supplemented by the use of a computer to evaluate solutions, however, research shows that such an approach compares quite favorably with more sophisticated computerized techniques.

Applying the Weighted-Distance Method

weighted-distance method
A mathematical model used to evaluate layouts (of facility locations) based on closeness factors.

When *relative* locations are a primary concern, such as for effective information flow, communication, material handling, and stockpicking, the weighted-distance method can be used to compare alternative block plans. The **weighted-distance method** is a mathematical model used to evaluate layouts based

on closeness factors. A similar approach, sometimes called the *load-distance method*, can be used to evaluate facility locations. The objective is to select a layout (or facility location) that minimizes the total weighted distances. The distance between two points is expressed by assigning the points to grid coordinates on a block diagram or map. An alternative approach is to use time rather than distance.

For a rough calculation, which is all that is needed for the weighted-distance method, either a Euclidean or rectilinear distance measure may be used. **Euclidean distance** is the straight-line distance, or shortest possible path, between two points. To calculate this distance, we create a graph. The distance between two points, say, points *A* and *B*, is

Euclidean distance
The straight-line distance, or shortest possible path, between two points.

$$d_{AB} = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2}$$

where

d_{AB} = distance between points *A* and *B*
x_A = *x*-coordinate of point *A*
y_A = *y*-coordinate of point *A*
x_B = *x*-coordinate of point *B*
y_B = *y*-coordinate of point *B*

Rectilinear distance measures the distance between two points with a series of 90-degree turns, as along city blocks. The distance traveled in the *x*-direction is the absolute value of the difference between the *x*-coordinates. Adding this result to the absolute value of the difference between the *y*-coordinates gives

rectilinear distance
The distance between two points with a series of 90-degree turns, as along city blocks.

$$d_{AB} = |x_A - x_B| + |y_A - y_B|$$

For assistance in calculating distances using either measure, see Tutor 3.1 in OM Explorer.

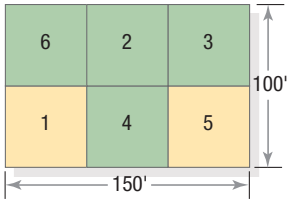
MyOMLab
Tutor 3.1 in MyOMLab provides an example to calculate both Euclidean and rectilinear distance measures.

The layout designer seeks to minimize the weighted-distance (*wd*) score by locating centers that have high-closeness ratings close together. To calculate a layout's *wd* score, we use either of the distance measures and simply multiply the proximity scores by the distances between centers. The sum of those products becomes the layout's final *wd* score—the lower the better. The location of a center is defined by its *x*-coordinate and *y*-coordinate.

EXAMPLE 3.1 Calculating the Weighted-Distance Score

The block plan in Figure 3.5 was developed using trial and error. A good place to start was to fix Departments 1 and 5 in their current locations. Then, the department pairs that had the largest closeness factors were located. The rest of the layout fell into place rather easily.

How much better, in terms of the *wd* score, is the proposed block plan shown in Figure 3.5 than the current plan shown in Figure 3.4? Use the rectilinear distance measure.



▲ **FIGURE 3.5**
Proposed Block Plan

MyOMLab
Active Model 3.1 in MyOMLab allows evaluation of the impact of swapping OBM departmental positions.

SOLUTION

The accompanying table lists each pair of departments that has a nonzero closeness factor in the closeness matrix. For the third column, calculate the rectilinear distances between the departments in the current layout. For example, departments 3 and 5 in the current plan are in the upper-left corner and bottom-right corner of the building, respectively. The distance between the centers of these blocks is three units (two horizontally and one vertically). For the fourth column, we multiply the weights (closeness factors) by the distances, and then add the results for a total *wd* score of 112 for the current plan. Similar calculations for the proposed plan produce a *wd* score of only 82. For example, between departments 3 and 5 is just one unit of distance (one vertically and zero horizontally).

		CURRENT PLAN		PROPOSED PLAN	
Department Pair	**Closeness Factor (*w*)**	**Distance (*d*)**	**Weighted-Distance Score (*wd*)**	**Distance (*d*)**	**Weighted-Distance Score (*wd*)**
1, 2	3	1	3	2	6
1, 3	6	1	6	3	18
1, 4	5	3	15	1	5
1, 5	6	2	12	2	12
1, 6	10	2	20	1	10
2, 3	8	2	16	1	8
2, 4	1	2	2	1	1

Current Plan

3	6	4
1	2	5

Proposed Plan

6	2	3
1	4	5

		CURRENT PLAN		PROPOSED PLAN	
Department Pair	Closeness Factor (*w*)	Distance (*d*)	Weighted-Distance Score (*wd*)	Distance (*d*)	Weighted-Distance Score (*wd*)
2, 5	1	1	1	2	2
3, 4	3	2	6	2	6
3, 5	9	3	27	1	9
4, 5	2	1	2	1	2
5, 6	1	2	2	3	3
			Total 112		Total 82

To be exact, we could multiply the two *wd* total scores by 50 because each unit of distance represents 50 feet. However, the relative difference between the two totals remains unchanged.

DECISION POINT

The *wd* score for the proposed layout makes a sizeable drop from 112 to 82, but management is not sure the improvement outweighs the cost of relocating four of the six departments (i.e., all departments but 1 and 5).

◉ Rectilinear Distances ○ Euclidean Distances

Department Pair	Closeness Factor	Distance	Score
1, 6	10	1	10
3, 5	9	1	9
2, 3	8	1	8
1, 3	6	1	6
1, 5	6	2	12
1, 4	5	3	15
1, 2	3	2	6
3, 4	3	2	6
4, 5	2	1	2
2, 4	1	1	1
2, 5	1	2	2
5, 6	1	3	3
		Total	80

6	2	4
1	3	5

▲ **FIGURE 3.6**
Second Proposed Block Plan (Analyzed with *Layout* Solver)

Although the *wd* score for the proposed layout in Example 3.1 represents an almost 27 percent improvement, the designer may be able to improve on this solution. Furthermore, the manager must determine whether the revised layout is worth the cost of relocating four of the six departments. If relocation costs are too high, a less-expensive proposal must be found.

OM Explorer and POM for Windows can help identify some even more attractive proposals. For example, one option is to modify the proposed plan by switching the locations of departments 3 and 4. OM Explorer's output in Figure 3.6 shows that the *wd* score for this second revision not only drops to 80, but requires that only three departments be relocated compared with the original layout in Figure 3.4. Perhaps this second proposed plan is the best solution.

Design a Detailed Layout

After finding a satisfactory block plan, the final step translates it into a detailed representation, showing the exact size and shape of each center; the arrangement of elements (e.g., desks, machines, and storage areas); and the location of aisles, stairways, and other service space. These visual representations can be two-dimensional drawings, three-dimensional models, or computer-aided graphics. This step helps decision makers discuss the proposal and problems that might otherwise be overlooked. Such visual representations can be particularly important when evaluating high customer-contact processes.

Customer Involvement

Having covered process structure decisions and how they are translated into a layout, we now turn to a second major decision—customer involvement—shown in Figure 3.1. Customer involvement reflects the ways in which customers become part of the process and the extent of their participation. It is especially important for many service processes, particularly if customer contact is (or should be) high.

While eBay devised one way to accommodate the variability created by customer involvement, Starbucks faces a different kind of customer variability. The coffee shop chain allows customers to choose among many permutations of sizes, flavors, and preparation techniques in its beverages.

In order to fill orders accurately and efficiently, Starbucks trains its counter clerks to call out orders to beverage makers in a particular sequence. It is even better when customers themselves can do so. Starbucks attempts to teach customers its ordering protocol. First, it provides a "guide-to-ordering pamphlet" for customers to look over. Second, it trains clerks to repeat the order in the correct sequence for the beverage makers, which may not be how the customer presented it. This process not only makes it easier for the beverage makers, but also indirectly "trains" the customers in how to place their orders.



ABB, Inc.

The detailed layout becomes a reality. Shown here is part of the office in one of the ABB facilities, a global leader in power and automation technologies. Here the workstations are small and semiprivate, but "outposts" are available. This common area has easy chairs fitted with arms that provide a surface for writing or laptops. People meet comfortably for face-to-face talks, rather than communicating by e-mail.

Possible Disadvantages

Customer involvement is not always a good idea. In some cases, giving the customer more active contact in a service process will just be disruptive, making the process less efficient. Managing the timing and volume of customer demands becomes more challenging if the customer is physically present and expects prompt delivery. Exposing the facilities and employees to the customer can have important quality implications (favorable or unfavorable). Such changes make interpersonal skills a prerequisite to the service provider's job, but higher skill levels come at a cost. Revising the facility layout might be a necessary investment, now that managing customer perceptions becomes an important part of the process. It also might mean having many smaller decentralized facilities closer to the various customer concentration areas if the customer comes to the service providers.

Possible Advantages

Despite these possible disadvantages, the advantages of a more customer-focused process might increase the net value to the customer. Some customers seek active participation in and control over the service process, particularly if they will enjoy savings in both price and time. The manager must assess whether advantages outweigh disadvantages, judging them in terms of the competitive priorities and customer satisfaction. More customer involvement can mean better quality, faster delivery, greater flexibility, and even lower cost. Self-service is the choice of many retailers, such as gasoline stations, supermarkets, and bank services. Manufacturers of products (such as toys, bicycles, and furniture) may also prefer to let the customer perform the final assembly because product, shipping, and inventory costs frequently are lower. Customer involvement can also help coordinate across the supply chain (see Chapter 12, "Integrating the Supply Chain"). Emerging technologies allow companies to engage in an active dialogue with customers and make them partners in creating value and forecasting future demand. Companies can also revise some of their traditional processes, such as pricing and billing systems, to account for their customers' new role. For example, in business-to-business relationships, the Internet changes the roles that companies play with other businesses. Suppliers to automobile companies can be close collaborators in the process of developing new vehicles and no longer are passive providers of materials and services. The same is true for distributors. Walmart does more than just distribute Procter & Gamble's products: It shares daily sales information and works with Procter & Gamble in managing inventories and warehousing operations.



Ramin Talaie/Corbis

A customer at Starbucks, a large coffee shop chain, places his order in the correct way. By structuring the ordering process for counter clerks and customers, Starbucks can deal efficiently with the variety in products offered, and with no hit on the service experience.



ABB, Inc.

Technicians in the parts repair department of this ABB facility must be flexible enough to repair many different parts for automation equipment installed at customer locations in the field. This operation has 30 different workstations configured to perform different types of processes. Workers are cross-trained to move from one station to another, depending on what needs to be done.

Resource Flexibility

Just as managers must account for customer contact when making customer involvement decisions, so must they account for process divergence and diverse process flows when making resource flexibility decisions in Figure 3.1. High task divergence and flexible process flows require more flexibility of the process's resources—its employees, facilities, and equipment. Employees need to perform a broad range of duties, and equipment must be general purpose. Otherwise, resource utilization will be too low for economical operations.

Workforce

Operations managers must decide whether to have a **flexible workforce**. Members of a flexible workforce are capable of doing many tasks, either at their own workstations or as they move from one workstation to another. However, such flexibility often comes at a cost, requiring greater skills and thus more training and education. Nevertheless, benefits can be large: Worker flexibility can be one of the best ways to achieve reliable customer service and alleviate capacity bottlenecks. Resource flexibility helps to absorb the feast-or-famine workloads in individual operations that are caused by low-volume production, divergent tasks, flexible flows, and fluid scheduling.

flexible workforce

A workforce whose members are capable of doing many tasks, either at their own workstations or as they move from one workstation to another.

The type of workforce required also depends on the need for volume flexibility. When conditions allow for a smooth, steady rate of output, the likely choice is a permanent workforce that expects regular full-time employment. If the process is subject to hourly, daily, or seasonal peaks and valleys in demand, the use of part-time or temporary employees to supplement a smaller core of full-time employees may be the best solution. However, this approach may not be practical if knowledge and skill requirements are too high for a temporary worker to grasp quickly.

Equipment

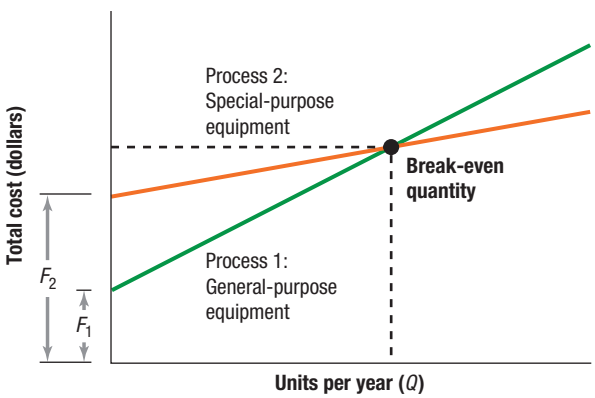
MyOMLab

Tutor 3.2 in MyOMLab demonstrates how to do break-even analysis for equipment selection.

Low volumes mean that process designers should select flexible, general-purpose equipment. Figure 3.7 illustrates this relationship by showing the total cost lines for two different types of equipment that can be chosen for a process. Each line represents the total annual cost of the process at different volume levels. It is the sum of fixed costs and variable costs (see Supplement A, "Decision Making Models"). When volumes are low (because customization is high), process 1 is the better choice. It calls for inexpensive general-purpose equipment, which keeps investment in equipment low and makes fixed costs (F_1) small. Its variable unit cost is high, which gives its total cost line a relatively steep slope. Process 1 does the job, but not at peak efficiency.

Conversely, process 2 is the better choice when volumes are high and customization is low. Its advantage is low variable unit cost, as reflected in the flatter total cost line. This efficiency is possible when customization is low because the equipment can be designed for a narrow range of products or tasks. Its disadvantage is high equipment investment and, thus, high fixed costs (F_2). When annual volume produced is high enough, spreading these fixed costs over more units produced, the advantage of low variable costs more than compensates for the high fixed costs.

▼ **FIGURE 3.7**
Relationship Between Process Costs and Product Volume



The break-even quantity in Figure 3.7 is the quantity at which the total costs for the two alternatives are equal. At quantities beyond this point, the cost of process 1 exceeds that of process 2. Unless the firm expects to sell more than the break-even amount, which is unlikely with high customization and low volume, the capital investment of process 2 is not warranted.

Capital Intensity

Capital intensity is the mix of equipment and human skills in the process; the greater the relative cost of equipment, the greater is the capital intensity. As the capabilities of technology increase and its costs decrease, managers face an ever-widening range of choices, from

operations utilizing very little automation to those requiring task-specific equipment and little human intervention. **Automation** is a system, process, or piece of equipment that is self-acting and self-regulating. Although automation is often thought to be necessary to gain competitive advantage, it has both advantages and disadvantages. Thus, the automation decision requires careful examination.

Automating Manufacturing Processes

Substituting labor-saving capital equipment and technology for labor has been a classic way of improving productivity and quality consistency in manufacturing processes. If investment costs are large, automation works best when volume is high, because more customization typically means reduced volume. Gillette, for example, spent $750 million on the production lines and robotics that gave it a capacity to make 1.2 billion razor cartridges a year. The equipment is complicated and expensive. Only with such high volumes could this line process produce the product at a price low enough that consumers could afford to buy it.

One big disadvantage of capital intensity can be the prohibitive investment cost for low-volume operations (see Figure 3.7). Generally, capital-intensive operations must have high utilization to be justifiable. Also, automation does not always align with a company's competitive priorities. If a firm offers a unique product or high-quality service, competitive priorities may indicate the need for hand labor and individual attention rather than new technology. A case in point is the downstream processes in Gillette's supply chain that package and store the razor cartridges. It customizes the packaging for different regions of the world, so that volumes for any one type of package are much lower. As a result of the low volumes, Gillette does not use expensive automation for these processes. In fact, it outsources them. Producing razor cartridges to stock using highly automated processes, and then packaging them in customized fashion at remote locations on demand, is also a good example of the principle of postponement.



LOOK Die Bildagentur der Fotografen GmbH/Alamy

Volkswagen aspires to become a full-line manufacturer of cars ranging from the smallest compacts to the largest luxury models. In the United States, for example, it offers the Jetta, GTI, Golf, Passat, CC, Routan, Tiguan, Touareg and Eos. The new VW Phaeton is the latest example of the brand's higher end. It uses a line process dedicated strictly to Phaeton cars. Because of its focus on Phaetons, which are made nowhere else, it enjoys high volumes. This volume justifies the high automation invested in this $208 million plant. The building is located in the heart of Dresden, a city known for its arts and craftsmanship. Its walls are made almost exclusively of glass, and it floors are covered entirely in Canadian maple. There are no smokestacks, no noises, and no toxic byproducts. Parts arrive and luxury cars depart. Of course, such a plant is not always possible, as with a steel mill or foundry. Shown in the photo is the arrival of the body structure that is painted in a plant about 60 miles from Dresden. It is what it looks like when it arrives, before the Phaeton assembly begins. For a photo tour, see **http://forums.vwvortex.com/showthread.php?1837641**.

Fixed Automation Manufacturers use two types of automation: (1) fixed and (2) flexible (or programmable). Particularly appropriate for line and continuous-flow process choices, **fixed automation** produces one type of part or product in a fixed sequence of simple operations. Operations managers favor fixed automation when demand volumes are high, product designs are stable, and product life cycles are long. These conditions compensate for the process's two primary drawbacks: (1) large initial investment cost and (2) relative inflexibility. However, fixed automation maximizes efficiency and yields the lowest variable cost per unit if volumes are high.

Flexible Automation **Flexible (or programmable) automation** can be changed easily to handle various products. The ability to reprogram machines is useful for both low-customization and high-customization processes. In the case of high customization, a machine that makes a variety of products in small batches can be programmed to alternate between products. When a machine has been dedicated to a particular product or family of products, as in the case of low customization and a line flow, and the product is at the end of its life cycle, the machine can simply be reprogrammed with a new sequence of tasks for a new product. An **industrial robot**, which is a versatile, computer-controlled machine programmed to perform various tasks, is a classic example of flexible automation. These "steel-collar" workers operate independently of human control. A robot's arm has up to six standard movements. The robot's "hand" actually does the work. The hand can be changed to perform different tasks, such as materials handling, assembly, and testing. **Managerial Practice 3.1** describes how the focus on operations efficiency at DP World have earned the company worldwide acclaim.

automation

A system, process, or piece of equipment that is self-acting and self-regulating.

fixed automation

A manufacturing process that produces one type of part or product in a fixed sequence of simple operations.

flexible (or programmable) automation

A manufacturing process that can be changed easily to handle various products.

industrial robot

Versatile, computer-controlled machine programmed to perform various tasks.

Automating Service Processes

Using capital inputs as a labor-saving device is also possible for service processes. In educational services, for example, long-distance learning technology now can supplement or even replace the traditional classroom experience by using books, computers, Web sites, and videos as facilitating goods that go with the service. Justifying technology need not be limited to cost reduction. Sometimes, it can actually allow more task divergence by making available a wide menu of choices to the customer.



Flexible Automation at Just Born, Inc., a candy company in Pennsylvania (**www.justborn.com**).

On the left, a robot picks up the PEEPS® brand marshmallow yellow bunnies with great speed in groups of four and places them into preformed trays, which then move to automatic shrink-wrapping machines. The trays are then boxed into cases for shipment. Upstream in this line process, the bunnies were extruded in shape on a belt with about 1/2 inch of sugar on it and then cooled as the conveyor moves along. The robot is regularly reprogrammed based on the marshmallow configuration being produced.

On the right, the robot using vacuum cups on the "hand" of its arm picks up boxes of five-pound MIKE AND IKE® bags that are packaged with six bags in a box. The robot reads the bar code, knows the pallet configuration, picks up and places the pallet and a thin cardboard liner on the conveyor, and stacks the boxes until complete. The pallet then comes toward the camera to be picked off the line by a fork truck, and then shrink-wrapped (automatically) and loaded into a 54-foot trailer. PEEPS® and MIKEANDIKE® are registered trademarks of Just Born, Inc. Used with permission.

PEEPS® and MIKE AND IKE® are registered trademarks of Just Born, Inc. Used with permission.

Technology in the future will surely make possible even a greater degree of customization and variety in services that currently only human providers can now deliver. Beyond cost and variety considerations, management must understand the customer and how much close contact is valued. If the customers seek a visible presence and personal attention, technologies reduced to sorting through a variety of options on the Internet or over the telephone might be a poor choice.

The need for volume to justify expensive automation is just as valid for service processes as for manufacturing processes. Increasing the volume lowers the cost per dollar of sales. Volume is essential for many capital-intensive processes in the transportation, communications, and utilities industries.

Economies of Scope

economies of scope

Economies that reflect the ability to produce multiple products more cheaply in combination than separately.

If capital intensity is high, resource flexibility usually is low. In certain types of manufacturing operations, such as machining and assembly, programmable automation breaks this inverse relationship between resource flexibility and capital intensity. It makes possible both high capital intensity and high resource flexibility, creating economies of scope. **Economies of scope** reflect the ability to produce multiple products more cheaply in combination than separately. In such situations, two conflicting competitive priorities—customization and low price—become more compatible. However, taking advantage of economies of scope requires that a family of parts or products have enough collective volume to utilize equipment fully.

Economies of scope also apply to service processes. Consider, for example, Disney's approach to the Internet. When the company's managers entered the volatile Internet world, their businesses were only weakly tied together. Disney's Infoseek business, in fact, was not even fully owned. However, once its Internet markets became more crystallized, managers at Disney moved to reap the benefits of economies of scope. They aggressively linked their Internet processes with one another and with other parts of Disney. A flexible technology that handles many services together can be less expensive than handling each one separately, particularly when the markets are not too volatile.

Strategic Fit

The manager should understand how the four major process decisions tie together, so as to spot ways of improving poorly designed processes. The choices should fit the situation and each other. When the fit is more *strategic*, the process will be more effective. We examine services and manufacturing processes, looking for ways to test for strategic fit.

MANAGERIAL PRACTICE 3.1 DP World Steps Up International Shipping

Originating in Dubai in 1999, the port management company Dubai Ports International (DPI) was initially formed to improve the efficiency of operations in Middle Eastern ports. After successfully transforming the Jeddah Islamic Port into the most successful in Saudi Arabia, other countries realized that DPI could improve their ports as well. In 2005, DPI combined with the Dubai Ports Authority and formed Dubai Ports World (DP World), which is now a world leader in international port management.

DP World's first major business move was the acquisition of CSX World Terminals in January 2005. This purchase gave the company a major presence in several Asian countries, Australia, Germany, Dominican Republic, and Venezuela. In addition to these already functioning ports, DP World also took over the contracts that CSX had for several major projects that were under construction. Since these mergers and acquisitions, DP World has continued to expand and currently holds contracts all over the world.

The business approach of DP World is customer centered, as they cater to the needs of individual ports. Their primary focus is to improve the efficiency of operations and, in doing so, to improve the profit margin of the port. These effective managerial practices have earned the company worldwide acclaim. A flat management structure is employed, with a focus on entrepreneurial drive rather than bureaucracy. These management practices have proved successful in turning around struggling ports and terminals time and time again.

With the purchase of six ports in the United States in 2006, DP World encountered a small scandal. The executive branch of the United States government approved the deal, but there were U.S. security concerns about a company based in the United Arab Emirates. U.S. authorities urged that the deal be completed, but legislation was introduced in Congress to delay the sale. The House of Representatives was vehemently opposed to the deal, and DP World ended up turning over management of the ports to a United States-based company. Despite this blip, DP World has continued to grow at an exponential rate.



© Corbis/Stephanie Kuykendal

Saudi Arabia recruited the help of Dubai Port International to regenerate the Jeddah Islamic port, which is now one of the most successful and thriving ports in the Middle East.

Dubai Ports World has attained international success through applying a simple managerial model to all of its business ventures. The company holds a number of major contracts for future ports and shows no signs of slowing down in the future. International trade is more prevalent than it has ever been in history, and DP World is a major force in improving port efficiency to aid the process of worldwide trade.

Source: "DP World Official Website," **www.dpworld.com**, Accessed September 29, 2008; "Congress Declares War on Ports Deal," Cnn.com, March 8, 2006, **http://www.cnn.com/2006/POLITICS/03/08/port.security/index.html.**

Decision Patterns for Service Processes

After analyzing a process and determining its position on the customer-contact matrix in Figure 3.2, it may be apparent that it is improperly positioned, either too far to the left or right, or too far to the top or bottom. Opportunities for improvement become apparent. Perhaps, more customization and customer contact is needed than the process currently provides. Perhaps, instead, the process is too divergent, with unnecessarily flexible flows. Reducing divergence might reduce costs and improve productivity.

The process should reflect its desired competitive priorities. Front offices generally emphasize top quality and customization, whereas back offices are more likely to emphasize low-cost operation, consistent quality, and on-time delivery. The process structure selected then points the way to appropriate choices on customer involvement, resource flexibility, and capital intensity. Figure 3.8 shows how these key process decisions are tied to customer contact. High customer contact at a front-office service process means:

1. *Process Structure.* The customer (internal or external) is present, actively involved, and receives personal attention. These conditions create processes with high divergence and flexible process flows.
2. *Customer Involvement.* When customer contact is high, customers are more likely to become part of the process. The service created for each customer is unique.
3. *Resource Flexibility.* High process divergence and flexible process flows fit with more flexibility from the process's resources—its workforce, facilities, and equipment.
4. *Capital Intensity.* When volume is higher, automation and capital intensity are more likely. Even though higher volume is usually assumed to be found in the back office, it is just as likely to be in the front office for financial services. Information technology is a major type of automation at many service processes, which brings together both resource flexibility and automation.

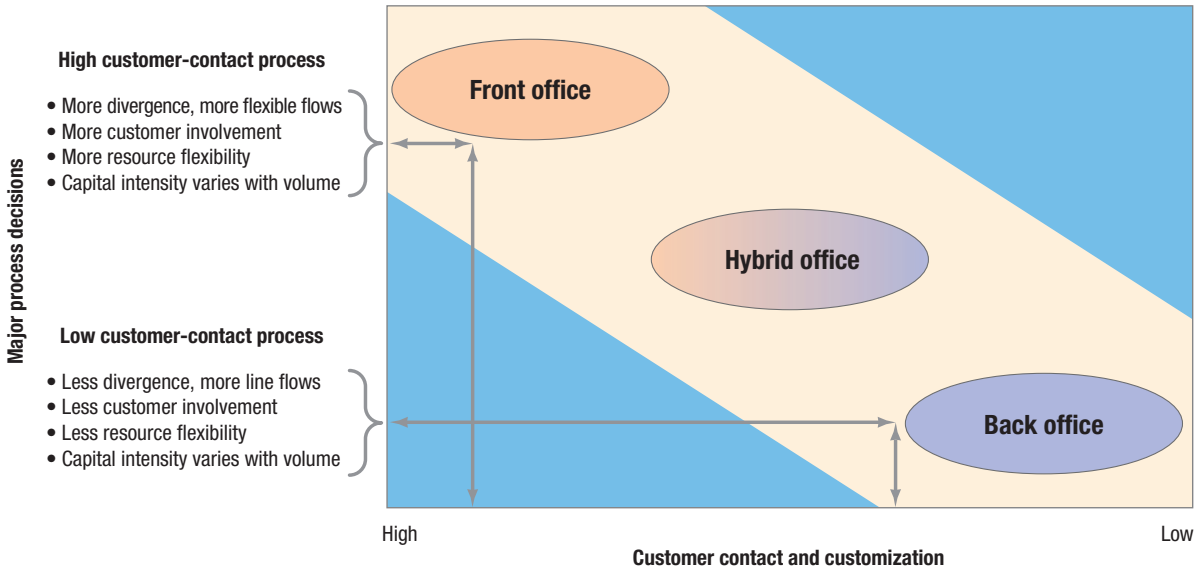


FIGURE 3.8 ▶
Decision Patterns for Service Processes

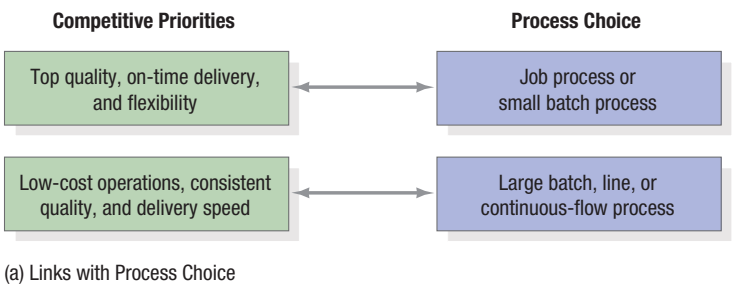
Of course, this list provides general tendencies rather than rigid prescriptions. Exceptions can be found, but these relationships provide a way of understanding how service process decisions can be linked coherently.

Decision Patterns for Manufacturing Processes

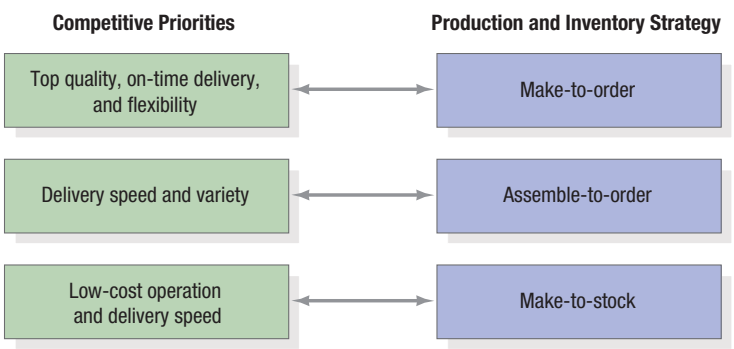
Just as a service process can be repositioned in the customer-contact matrix, a manufacturing process can also be moved in the product-process matrix. Changes can be made either in the horizontal direction of Figure 3.3 by changing the degree of customization and volume, or they can be moved in the vertical direction by changing process divergence. The production and inventory strategy can also be changed. Competitive priorities must be considered when translating strategy into specific manufacturing processes. Figure 3.9 shows some usual tendencies found in practice. Job and small batch processes are usual choices if top quality, on-time delivery, and flexibility (customization, variety, and volume flexibility) are given primary emphasis. Large batch, line, and continuous-flow processes match up with an emphasis on low-cost operations, consistent quality, and delivery speed.

For production and inventory strategies, the make-to-order strategy matches up with flexibility (particularly customization) and top quality. Because delivery speed is more difficult, meeting due dates and on-time delivery get the emphasis on the time dimension. The assemble-to-order strategy allows delivery speed and flexibility (particularly variety) to be achieved, whereas the make-to-stock strategy is the usual choice if delivery speed and low-cost operations are emphasized. Keeping an item in stock assures quick delivery because it is generally available when needed, without delays in producing it. High volumes open up opportunities to reduce costs.

▼ FIGURE 3.9
Links of Competitive Priorities with Manufacturing Strategy



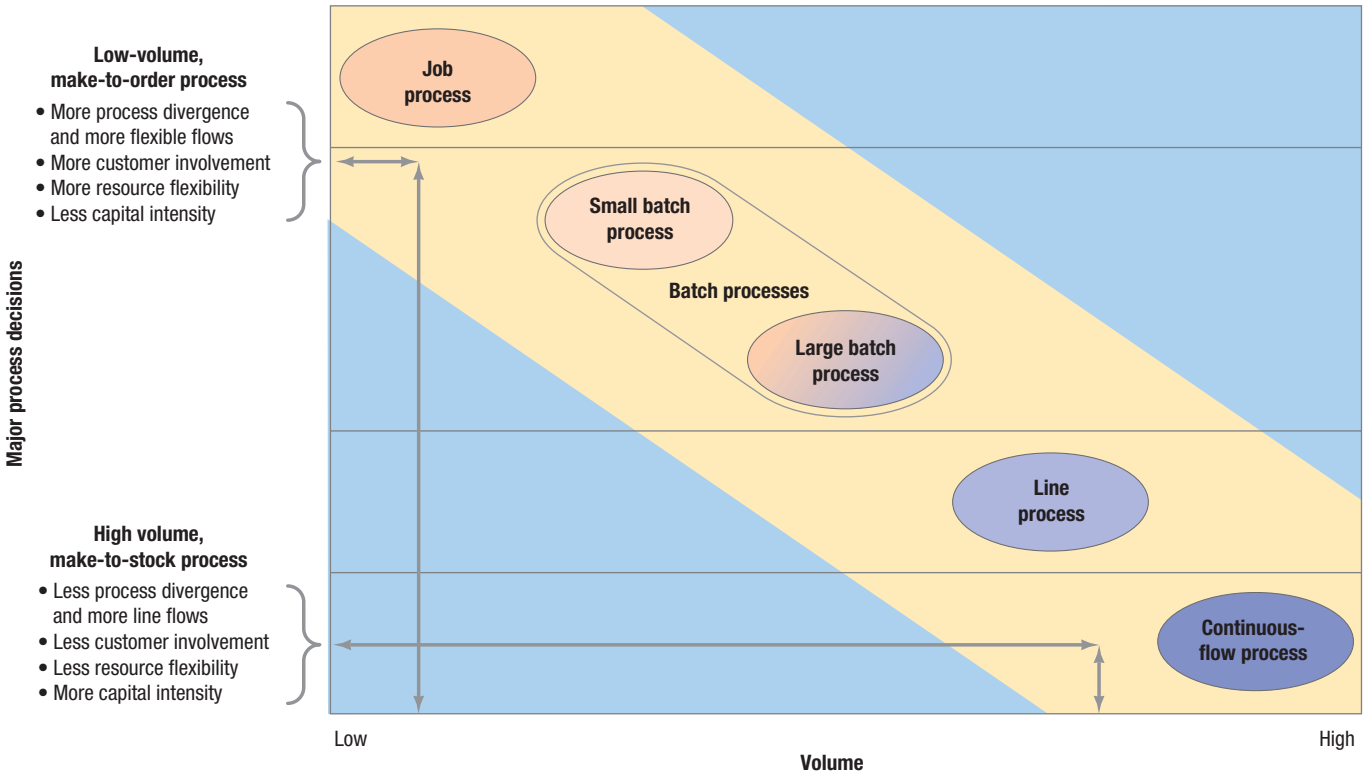
(a) Links with Process Choice



(b) Links with Production and Inventory Strategy

The process structure selected once again points the way to appropriate choices on customer involvement, resource flexibility, and capital intensity. Figure 3.10 summarizes the relationships between volume and the four key process decisions. High volumes per part type at a manufacturing process typically mean:

1. *Process Structure.* High volumes, combined with a standard product, make a line flow possible. It is just the opposite where a job process produces to specific customer orders.
2. *Customer Involvement.* Customer involvement is not a factor in most manufacturing processes, except for choices made on product variety and customization. Less discretion is allowed with line or continuous-flow processes in order to avoid the unpredictable demands required by customized orders.



▲ **FIGURE 3.10**
Decision Patterns for Manufacturing Processes

3. *Resource Flexibility.* When volumes are high and process divergence is low, flexibility is not needed to utilize resources effectively, and specialization can lead to more efficient processes.
4. *Capital Intensity.* High volumes justify the large fixed costs of an efficient operation. The King Soopers's bread line (see *The Big Picture* and the video in MyOMLab) is capital-intensive. It is automated from dough mixing to placement of the product on shipping racks. Expanding this process would be expensive. By way of contrast, the King Soopers's custom cake process is labor-intensive and requires little investment to equip the workers.

MyOMLab

Gaining Focus

In the past, new services or products often were added to a facility in the name of better utilizing fixed costs and keeping everything under the same roof. The result was a jumble of competitive priorities, process structures, and technologies. In the effort to do everything, nothing was done well.

Focus by Process Segments A facility's operations often can neither be characterized nor actually designed for one set of competitive priorities and one process choice. King Soopers (see *The Big Picture* and video in MyOMLab) had three processes under one roof, but management segmented them into three separate operations that were relatively autonomous. At a services facility, some parts of the process might seem like a front office and other parts like a back office. Such arrangements can be effective, provided that sufficient focus is given to each process.

MyOMLab

Plants within plants (PWPs) are different operations within a facility with individualized competitive priorities, processes, and workforces under the same roof. Boundaries for PWPs may be established by physically separating subunits or simply by revising organizational relationships. At each PWP, customization, capital intensity volume, and other relationships are crucial and must be complementary. The advantages of PWPs are fewer layers of management, greater ability to rely on team problem solving, and shorter lines of communication between departments.

plants within plants (PWPs)
Different operations within a facility with individualized competitive priorities, processes, and workforces under the same roof.

Focused Service Operations Service industries also implement the concepts of focus and PWPs. Specialty retailers opened stores with smaller, more accessible spaces. These focused facilities generally chipped away at the business of large department stores. Using the same philosophy,



AP Photo/Mike Wintroath

Focused factories are not just found in manufacturing. This single-speciality facility focuses just on heart surgery and has all of the advanced resources needed that cannot be provided by a general hospital. Another example is the Toronto-based Shouldice Clinic, which focuses just on hernias.

some department stores now focus on specific customers or products. Remodeled stores create the effect of many small boutiques under one roof.

Focused Factories Hewlett-Packard, Rolls-Royce, Japan's Ricoh and Mitsubishi, and Britain's Imperial Chemical Industries PLC are some of the firms that created **focused factories**, splitting large plants that produced all the company's products into several specialized smaller plants. The theory is that narrowing the range of demands on a facility will lead to better performance because management can concentrate on fewer tasks and lead a workforce toward a single goal.

Strategies for Change

The four major process decisions represent broad, strategic issues. Decisions that are made must be translated into actual process designs or redesigns. We conclude with two different but complementary philosophies for process design: (1) process reengineering and (2) process improvement.

Process Reengineering

Reengineering is the fundamental rethinking and radical redesign of processes to improve performance dramatically in terms of cost, quality, service, and speed. Process reengineering is about reinvention, rather than incremental improvement. It is strong medicine and not always needed or successful. Pain, in the form of layoffs and large cash outflows for investments in information technology, almost always accompanies massive change. However, reengineering processes can have big payoffs. Table 3.4 lists the key elements of the overall approach.

Reengineering has led to many successes and will continue to do so. However, it is not simple or easily done, nor is it appropriate for all processes or all organizations. The best understanding of a process, and how to improve it, often lies with the people who perform the work each day, not with cross-functional teams or top management.

focused factories

The result of a firm's splitting large plants that produced all the company's products into several specialized smaller plants.

reengineering

The fundamental rethinking and radical redesign of processes to improve performance dramatically in terms of cost, quality, service, and speed.

TABLE 3.4 | KEY ELEMENTS OF REENGINEERING

Element	Description
Critical processes	The emphasis of reengineering should be on core business processes. Normal process-improvement activities can be continued with the other processes.
Strong leadership	Senior executives must provide strong leadership for reengineering to be successful. Otherwise, cynicism, resistance ("we tried that before"), and boundaries between departments can block radical changes.
Cross-functional teams	A team, consisting of members from each functional area affected by the process change, is charged with carrying out a reengineering project. Self-managing teams and employee empowerment are the rule rather than the exception.
Information technology	Information technology is a primary enabler of process engineering. Most reengineering projects design processes around information flows, such as customer order fulfillment.
Clean-slate philosophy	Reengineering requires a "clean-slate" philosophy—that is, starting with the way the customer wants to deal with the company. To ensure a customer orientation, teams begin with internal and external customer objectives for the process.
Process analysis	Despite the clean-slate philosophy, a reengineering team must understand things about the current process: what it does, how well it performs, and what factors affect it. The team must look at every procedure involved in the process throughout the organization.

Process Improvement

Process improvement is the systematic study of the activities and flows of each process to improve it. Its purpose is to "learn the numbers," understand the process, and dig out the details. Once a process is really understood, it can be improved. The relentless pressure to provide better quality at a lower price means that companies must continually review all aspects of their operations. Process improvement goes on, whether or not a process is reengineered. There is always a better way.

process improvement
The systematic study of the activities and flows of each process to improve it.

An individual or a whole team examines the process, using the tools described in the next chapter. One must look for ways to streamline tasks, eliminate whole processes entirely, cut expensive materials or services, improve the environment, or make jobs safer. One must find the ways to trim costs and delays and to improve customer satisfaction.

LEARNING GOALS IN REVIEW

1. **Explain why processes exist everywhere in all organizations.** The "Process Strategy across the Organization" section, pp. 110–111, demonstrates that supply chain process exist throughout the supply chain and actually exist in all places throughout the organization. Pay particular attention to Tables 3.1 and 3.2.
2. **Discuss the four major process decisions.** The "Process Strategy Decisions" section, pp. 111–112, identifies the four key decisions around which we develop a vocabulary for understanding operations. Figure 3.1 shows how they interact in creating an effective process design.
3. **Position a process on the customer-contact matrix or product-process matrix.** The Customer-Contact Matrix for service processes in Figure 3.2 shows how the degree of customer contact and customization are linked with process divergence and line flows. Three natural positions emerge: the front office, hybrid office, and back office. In manufacturing, the key drivers are customization and volume, which are linked with line flows and the extent of repetitive work. Figure 3.3 shows these relationships in the form of the Product-Process Matrix, with natural positions ranging from the job process to the continuous flow process.
4. **Configure operations into layouts.** The "Layout" section, pp. 117–120, puts the process structure into a physical form by showing here each operation is located within the facility. Example 3.1 shows how to develop a block plan, and evaluate it with the help of the *Layout* Solver of OM Explorer. See Solved Problem 3.1 for another example.
5. **Define customer involvement, resource flexibility, capital intensity, and economies of scope.** These topics are covered in the "Customer Involvement," "Resource Flexibility," "Capital Intensity," and "Economies of Scope" sections on pp. 120–124. Note that customer involvement has advantages and disadvantages, resource flexibility applies to both workforce and equipment, and economies of scope in certain situations can break the inverse relationship between resource flexibility and capital intensity.
6. **Discuss how process decisions should fit together.** The "Strategic Fit" section, pp. 124–128, describes how the four major process decisions should tie together. Figures 3.8 and 3.9 show the decision patterns in pictorial form. The section concludes with a way to achieve these patterns by gaining focus, either with focused factories or gaining focus by process segments.
7. **Define process reengineering and process improvement.** The "Strategies for Change," pp. 128–129, describe both approaches to finding better process designs. Table 3.4 gives the key elements of reengineering. Process improvement is more of an incremental approach which uses the tools described in the next chapter.

MyOMLab helps you develop analytic skills and assesses your progress with multiple problems on the break-even analysis in choosing between two different processes, the weighted-distance method, and layout.

MyOMLab Resources	Titles	Link to the Book
Videos	*King Soopers Bakery: Process Choice* *Process Choice: Pearson Education Information Technology*	Manufacturing Process Structuring Capital Intensity
OM Explorer Solver	Layout	Layout
OM Explorer Tutors	3.1 Distance Measures 3.2 Breakeven for Equipment Selection	Applying the Weighted-Distance Method Example 3.1 (pp. 119–120) Resource Flexibility Figure 3.7 (p. 122)
POM for Windows: Layout	Layout	Example 3.1 (pp. 119–120)

MyOMLab Resources	Titles	Link to the Book
Tutor Exercises	3.1—Mt. Mudge 3.2—Break-Even for Equipment Selection	Applying the Weighted-Distance Method Example 3.1 (pp. 119–120) Resource Flexibility Figure 3.7 (p. 122)
SmartDraw	Often used to prepare detailed layouts and floor plans	Detailed Layout
Virtual Tours	1. Leannie Company Doll Factory 2. LA Aluminum Casting Company	Process Choice; Production and Inventory Strategy Process Choice; Production and Inventory Strategy
Internet Exercises	3.1—United Parcel Service 3.2—Carnival and Twilight 3.3—Timbuk2	Service Strategy; Customer Involvement
Advanced Problems	3.1 CCI Electronics 3.2 Getwell Hospital	Layout Layout
Additional Cases	Car Lube Operations Hightech, Inc. The Pizza Connection Bill's Hardware The Big Picture: Process Choice at King Soopers Bakery	Layout Layout Design a Detailed layout Capital Intensity Manufacturing Process Structuring
Key Equations		
Image Library		

Key Equations

1. Euclidean distance: $d_{AB} = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2}$
2. Rectilinear distance: $d_{AB} = |x_A - x_B| + |y_A - y_B|$

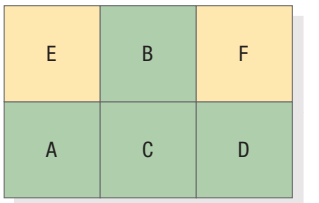
Key Terms

assemble-to-order strategy 116
automation 123
back office 114
batch process 116
block plan 118
capital intensity 112
closeness matrix 118
continuous flow process 116
customer contact 113
customer involvement 112
economies of scope 124
Euclidean distance 119
fixed automation 123
flexible (or programmable) automation 123
flexible flow 113
flexible workforce 122
focused factories 128
front office 114
hybrid office 114
industrial robot 123
job process 115
layout 112
line flow 114
line process 116
make-to-order strategy 116
make-to-stock strategy 117
mass customization 117
mass production 117
plants within plants (PWPs) 127
postponement 116
process choice 115
process divergence 113
process improvement 129
process strategy 110
process structure 112
rectilinear distance 119
reengineering 128
resource flexibility 112
supply chain processes 111
weighted-distance method 118

Solved Problem 1

A defense contractor is evaluating its machine shop's current layout. Figure 3.11 shows the current layout, and the table shows the closeness matrix for the facility measured as the number of trips per day between department pairs. Safety and health regulations require departments E and F to remain at their current locations.

	TRIPS BETWEEN DEPARTMENTS					
Department	**A**	**B**	**C**	**D**	**E**	**F**
A	—	8	3		9	5
B		—		3		
C			—		8	9
D				—		3
E					—	3
F						—



▲ **FIGURE 3.11**
Current Layout

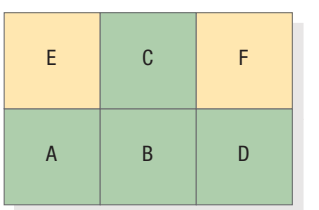
a. How much better is your layout than the current layout in terms of the *wd* score? Use rectilinear distance.

b. Use trial and error to find a better layout.

SOLUTION

a. The table reveals that the *wd* score drops from 92 for the current plan to 67 for the revised plan, a 27 percent reduction.

b. In addition to keeping departments E and F at their current locations, a good plan would locate the following department pairs close to each other: A and E, C and F, A and B, and C and E. Figure 3.12 was worked out by trial and error and satisfies all these requirements. Start by placing E and F at their current locations. Then, because C must be as close as possible to both E and F, put C between them. Place A below E, and B next to A. All of the heavy traffic concerns have now been accommodated. Department D, located in the remaining space, does not need to be relocated.



▲ **FIGURE 3.12**
Proposed Layout

		CURRENT PLAN		PROPOSED PLAN	
Department Pair	**Number of Trips (1)**	**Distance (2)**	***wd* Score (1) × (2)**	**Distance (3)**	***wd* Score (1) × (3)**
A, B	8	2	16	1	8
A, C	3	1	3	2	6
A, E	9	1	9	1	9
A, F	5	3	15	3	15
B, D	3	2	6	1	3
C, E	8	2	16	1	8
C, F	9	2	18	1	9
D, F	3	1	3	1	3
E, F	3	2	6	2	6
			wd = 92		*wd* = 67

Discussion Questions

1. Consider the range of processes in the financial services industry. What position on the customer-contact matrix would the process of selling financial services to municipalities occupy? The process of preparing monthly fund balance reports? Explain why they would differ.
2. Rate operators at a call center, who respond to queries from customers who call in about the company's product, on each of the five dimensions of customer contact in Table 3.3. Use a seven-point scale, where 1 = very low and 7 = very high. For example, the operators newer are physically present with the customer, and so they would get a score of 1 for physical presence. Explain your ratings, and then calculate a combined score for the overall customer contact. Did you use equal weights in calculating the combined score? Why or why not? Where is your process positioned on the customer-contact matrix? Is it properly aligned? Why or why not?
3. How do the process strategies of eBay and McDonald's differ, and how do their choices relate to customer-introduced variability?
4. Medical technology can outfit a patient with an artificial heart, or cure vision defects with the touch of a laser. However, hospitals still struggle with their back-office processes, such as getting X-ray files from radiology on the fourth

floor to the first-floor view boxes in the emergency room without having to send a runner. More than 90 percent of the estimated 30 billion health transactions each year are conducted by telephone, fax, or mail. To what extent, and how, can information technology improve productivity and quality for such processes? Remember that some doctors are not ready to give up their pads and pencils, and many hospitals have strong lines drawn around its departments, such as pharmacy, cardiology, radiology, and pediatrics.

5. What processes at manufacturing firms are really service processes that involve considerable customer contact? Can customer contact be high, even if the process only has internal customers?
6. Performance criteria important in creating a layout can go well beyond communication and materials handling. Identify the types of layout performance criteria that might be most important in the following settings.
   a. Airport
   b. Bank
   c. Classroom
   d. Product designers' office
   e. Law firm
7. Consider this sign seen in a local restaurant: "To-go orders do NOT include complimentary chips and salsa. If you have any questions, see our management, NOT our employees." What impact does this message have on its employees, their service processes, and customer satisfaction? Contrast this approach with the one taken by a five-star restaurant. Are the differences primarily due to different competitive priorities?
8. Select one of the three processes shown in the MyOMLab's video for King Soopers (bread, pastry, or custom cakes). What kind of transformation process, process choice, and inventory strategy are involved? Is the process properly aligned? Explain.

Problems

The OM Explorer and POM for Windows software is available to all students using the 10th edition of this textbook. Go to **www.pearsonglobaleditions.com/krajewski** to download these computer packages. If you purchased MyOMLab, you also have access to Active Models software and significant help in doing the following problems. Check with your instructor on how best to use these resources. In many cases, the instructor wants you to understand how to do the calculations by hand. At the least, the software provides a check on your calculations. When calculations are particularly complex and the goal is interpreting the results in making decision, the software entirely replaces the manual calculations.

Problems 1 and 2 apply break-even analysis (discussed in Supplement A, "Decision Making Models") to process decisions.

1. Two different manufacturing processes are being considered for making a new product. The first process is less capital-intensive, with fixed costs of only $50,000 per year and variable costs of $700 per unit. The second process has fixed costs of $400,000, but has variable costs of only $200 per unit.
   a. What is the break-even quantity, beyond which the second process becomes more attractive than the first?
   b. If the expected annual sales for the product is 800 units, which process would you choose?
2. Dr. Gulakowicz is an orthodontist. She estimates that adding two new chairs will increase fixed costs by $147,000, including the annual equivalent cost of the capital investment and the salary of one more technician. Each new patient is expected to bring in $3,919 per year in additional revenue, with variable costs estimated at $1,030 per patient. The two new chairs will allow Dr. Gulakowicz to expand her practice by as many as 195 patients annually. How many patients would have to be added for the new process to break even?
3. Baker Machine Company is a job shop that specializes in precision parts for firms in the aerospace industry. Figure 3.13 shows the current block plan for the key manufacturing centers of the 75,000-square-foot facility. Refer to the following closeness matrix and use rectilinear distance (the current distance from inspection to shipping and receiving is three units) to calculate the change in the weighted distance, *wd*, score if Baker exchanges the locations of the tool crib and inspection.

CLOSENESS MATRIX

	Trips Between Departments					
Department	**1**	**2**	**3**	**4**	**5**	**6**
1. Burr and grind	—	8	3		10	6
2. Numerically controlled (NC) equipment		—		2		
3. Shipping and receiving			—		9	10
4. Lathes and drills				—		3
5. Tool crib					—	4
6. Inspection						—

3	4	2
1	5	6

◀ **FIGURE 3.13** Current Layout

4. Baker Machine (see Problem 3) is considering two alternative layouts. Compare the *wd* scores using rectilinear distance of the following two block plans to determine which alternative layout is better.

3	6	4
5	1	2

◀ **FIGURE 3.13(a)** Alternative Layout 1

3	1	4
5	6	2

◀ **FIGURE 3.13(b)**
Alternative Layout 2

5. The head of the information systems group at Conway Consulting must assign six new analysts to offices. The following closeness matrix shows the expected frequency of contact between analysts. The block plan in Figure 3.14 shows the available office locations (1–6) for the six analysts (A–F). Assume equal-sized offices and rectilinear distance.

CLOSENESS MATRIX

	Contacts Between Analysts					
Analyst	**A**	**B**	**C**	**D**	**E**	**F**
Analyst A	—		6			
Analyst B		—		12		
Analyst C			—	2	7	
Analyst D				—		4
Analyst E					—	
Analyst F						—

1	2	3
4	5	6

◀ **FIGURE 3.14**
Conway Consulting's Block Plan

Evaluate the *wd* scores of the following three alternative layouts, again assuming rectilinear distance, and determine which is best.

B	C	D
A	E	F

◀ **FIGURE 3.14(a)**
Alternative Layout 1

C	B	D
A	F	E

◀ **FIGURE 3.14(b)**
Alternative Layout 2

F	B	D
A	C	E

◀ **FIGURE 3.14(c)**
Alternative Layout 3

6. The department of engineering at a university in New Jersey must assign six faculty members to their new offices. The following closeness matrix indicates the expected number of contacts per day between professors. The available office spaces (1–6) for the six faculty members are shown in Figure 3.15. Assume equal-sized offices. The distance between offices 1 and 2 (and between offices 1 and 3) is 1 unit, whereas the distance between offices 1 and 4 is 2 units.

CLOSENESS MATRIX

	Contacts Between Professors					
Professor	**A**	**B**	**C**	**D**	**E**	**F**
A	—		4			
B		—		12		10
C			—	2	7	
D				—		4
E					—	
F						—

a. Because of their academic positions, Professor A must be assigned to office 1, Professor C must be assigned to office 2, and Professor D must be assigned to office 6. Which faculty members should be assigned to offices 3, 4, and 5, respectively, to minimize the total weighted-distance score (assuming rectilinear distance)?

b. What is the weighted-distance score of your solution?

1	2
3	4
5	6

◀ **FIGURE 3.15**
Available Space

7. Richard Garber is the head designer for Matthews and Novak Design Company. Garber has been called in to design the layout for a newly constructed office building. From statistical samplings over the past three months, Garber developed the following closeness matrix for daily trips between the department's offices.

CLOSENESS MATRIX

	Trips Between Departments					
Department	**A**	**B**	**C**	**D**	**E**	**F**
A	—	25	90			185
B		—			105	
C			—		125	125
D				—	25	
E					—	105
F						—

C	F	A
B	E	D

◀ **FIGURE 3.16**
Alternative Block Plan

a. If other factors are equal, which two offices should be located closest together?

b. Figure 3.16 shows an alternative layout for the department. What is the total weighted-distance score for this plan based on rectilinear distance and assuming that offices A and B are three units of distance apart?

c. Use the explicit enumeration method of the POM for Windows software to find the block plan that minimize the total weighted-distance score.

8. A firm with four departments has the following closeness matrix and the current block plan shown in Figure 3.17.

a. What is the weighted-distance score for the current layout (assuming rectilinear distance)?

CLOSENESS MATRIX

	Trips Between Departments			
Department	**A**	**B**	**C**	**D**
A	—	12	10	8
B		—	20	6
C			—	0
D				—

A	B
C	D

◀ **FIGURE 3.17**
Current Block Plan

b. Develop a better layout. What is its total weighted-distance score?

Active Model Exercise

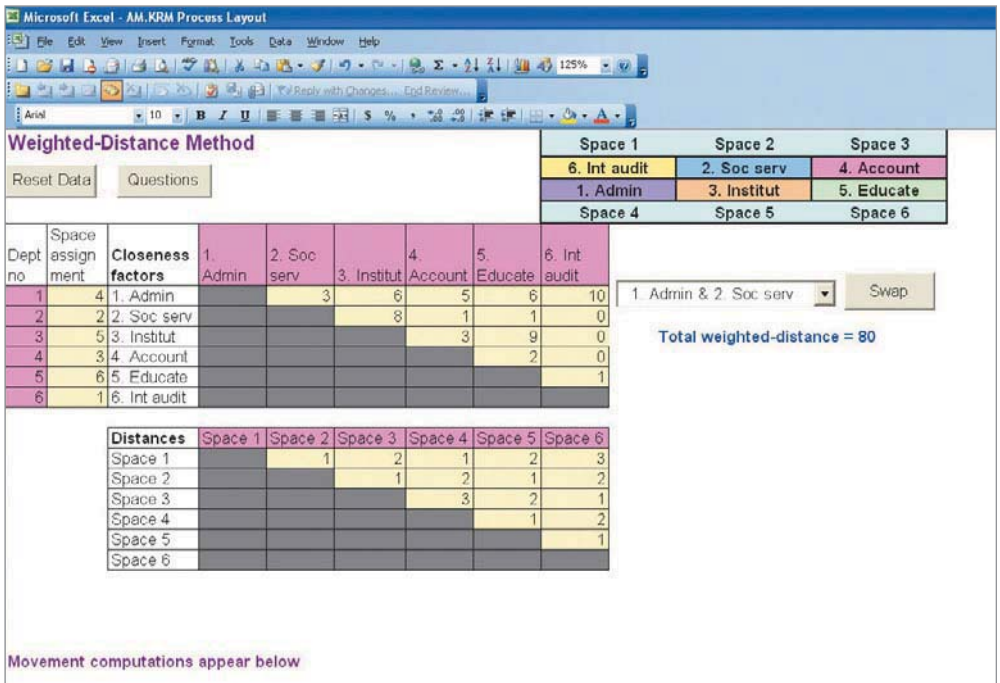
This Active Model for Example 3.1 appears in MyOMLab. It allows you to see the effects of performing paired swaps of departments.

QUESTIONS

1. Use the swap button one swap at a time. If the swap helps, move to the next pair. If the swap does not help, hit the swap button once again to put the departments back. What is the minimum weighted-distance score after all swaps have been tried?

2. Look at the two data tables, and use the yellow-shaded column to put departments in spaces. What space assignments lead to the minimum cost? What is this cost?

3. What is the current total weighted-distance score?



Layout Using Data from Example 3.1

CASE Sims Metal Management

Background

Sims Metal Management is the world's largest metal recycler and has operations over the five continents with more than 90 years of experience in the business, which started in Australia. Every year they process over 2 million metric tons of ferrous and nonferrous materials in the United Kingdom across 36 metal recycling locations. The company recovers end-of-life and abandoned cars, old machinery, nonhazardous consumer goods, even trains. There are growing concerns in the manufacturing world over the global supply chain of commodities. Some materials such as rare earth elements are becoming increasingly expensive due to a restraint of their export from China and an ever-increasing demand from the electronic industry. Furthermore, the sustained demand from BRIC countries (Brazil, Russia, India, and China, all developing economies, but with tremendous potential thanks to their territories and increasing presence in world-trade) has raised commodity prices on iron ore, steel, and oil, while only naming the most common. Experts agree that the world has entered an upward trend in commodity prices at the beginning of the century and it is identified as a supercycle projected to last 30 years.

Recycling as a sustainable business practice

Governments and large industries are starting to look into recycling end-of-life products, ranging from cars to cell phones to retrieve these materials and their strategic supply chains are evolving accordingly. Sims Metal Management and other recyclers are, no doubt, on a growing market trend. In addition to the economic laws of supply and demand, the European Union, in its bid to become the most environmentally friendly economic group, has issued directives and laws to channel the disposal of hazardous materials from landfills and make compulsory the use of specialized companies to dismantle old factories.

Sims Metal Management has become the largest world recycler thanks to a series of acquisitions over the past years and notably in the United Kingdom in 2002. The competition is fierce and Sims proposes competitive prices on nonferrous metals (aluminum, brass, and copper) thanks to high recovery rates. These can be achieved owing to an efficient supply chain and well-designed internal processes.

SIMS Metal strategy

The company's strategy, to stay the number one metals recycler in the UK, relies on one fundamental principle: low operating costs with high recycling rates to offer the best quotes when bidding to recycle metals, when decommissioning a plant for example. This strategy is achieved thanks to economies of scales, as for any manufacturing business, and thanks to their recycling processes. To keep its operating costs to minimum, the company has sought, and received, the ISO 9000:2001 certification. The certification focuses on Quality Management, with the aim to satisfy customers but also to continually improve processes to sustain productivity gains, hence their competitive advantage. The operations of SIMS Metal ensure cars are shredded in a minimum amount of time. The company also minimises the amount of manual labor to keep its operating costs down. However, final steps of the shredding process need manual labour to finalise materials separation. The recycling industry is increasingly looking into manufacturing techniques to improve their operations. The car manufacturing industry has many methods that can be adapted to the recycling business: Lean, Six Sigma, Control Charts, etc. All these techniques have the common aim to reduce overall lead times and sustain productivity gains, thus serving the company strategy to remain price competitive.

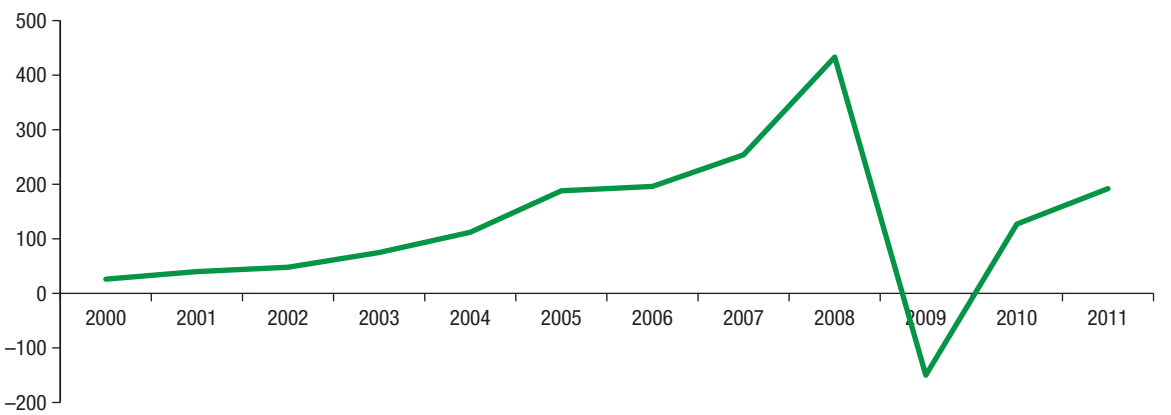
Metal shredding process

One of their most impressive processes is the treatment of end-of-life of vehicles (ELV). The cars are collected thanks to their transportation network linking the company 36 metals and shredding recycling locations across England and Wales to a network of 35 scrap metal dealers. The main difference between SIMS and other recyclers in the UK is their integrated operations.

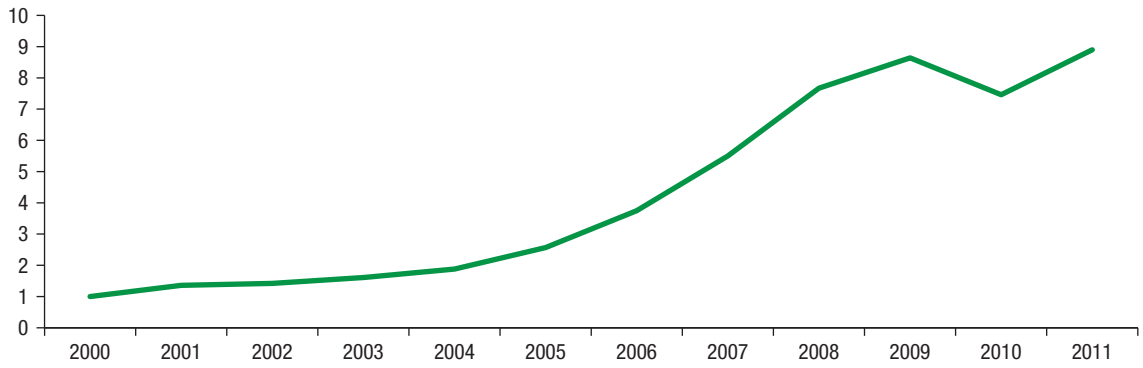
With their network of scrap dealers and delivery trucks, they ensure the constant flow of materials to the recycling facilities. Once metal is collected, it has to be transformed into a value-added product as fast as possible, otherwise it is only waste, without any value. For recyclers value is created with metals separation and cleaning.

To ensure the highest environmental standards, the cars are first depolluted. They are entered in the database and the owner is issued a certificate of destruction. Wheels, battery, bumpers, seats, and exhaust pipe are removed. Some of these elements, if still of good quality, will then be resold as secondhand goods. The car is then elevated and a technician safely drains fuel, waste oil, brake fluid, and coolant from the different tanks. Having been stripped of its major pollutants, the shredding process can start.

The car is loaded on a large conveyor and goes through two flattening wheels. The reduced metal is shredded in hundreds of parts containing



▲ **FIGURE 3.18**
SIMS Metal Management Global profits after tax in million US$



▲ **FIGURE 3.19**
SIMS Metal Management Global revenue in billion US$

ferrous, and nonferrous materials, but also plastic. The shredded elements pass through a chamber equipped with a powerful wind turbine to remove light materials (foam and cloth, for example). The remaining heavy materials pass in front of powerful magnets to separate ferrous and nonferrous components. Operatives now remove any nonferrous product that might not have been separated in the previous stage.

Electronic waste recycling process

In addition to recycling metal, Sims is recycling more and more electronic and electrical waste.

The European Union is restricting the disposal of electronic waste in landfills as batteries contain mercury and acids. Today in the United Kingdom, there are more than 50 million cell phones that have an average life of two years. To follow the consuming trend, Sims Metal Management offers an electronic waste recycling service for PCs and cell phones, but also for white goods such as refrigerators and washing machines.

Electronic waste is loaded on a conveyor where operatives separate batteries and large copper elements. A first reduction process occurs with the electronic waste being shredded into parts not exceeding 100 mm, hence preparing the materials for the secondary process. The material is then dropped on a hopper to evenly spread it across the width of the conveyor belt. This avoids having agglomerates of materials before entering the recycling machines. The materials go through another shredder to further reduce their size. This step is very polluting and dust is captured with a wind turbine. While still on the conveyor belt, the ferrous and nonferrous elements are separated with a large magnet, as previously described. The material is now composed of nonferrous elements and plastic. Even if deemed nonferrous, the material still has a marginal share of elements responding to magnetic fields. The material goes through an Eddy current (by alternating magnetic fields) and is separated between nonferrous and plastics. The separated material can now be resold to smelters.

The future of recycling?

The UK operations of SIMS Metal Management were acquired in 2002. Figure 3.18 highlights the company's profits which are linked to the efficiency of their operations, whilst Figure 3.19 underlines a constant revenue increase through acquisitions. Due to external growth with series of acquisitions, there are disparities across the organisation across several continents, for example SIMS Metal in the USA has facilities to specifically recycle aerospace metals. Also the management of electronic waste is different from one continent from another. For example in the UK, the European Union directives prevail whereas in the USA, it is still regulated by federal laws, not mentioning the differences in electronic waste legislation between the USA and Europe. With a global recycling network and taking advantage of its specialised facilities, SIMS Metal could open the door to specialised recycling, with countries giving advantage to certain types of recycling, as it has happened in manufacturing during the last 30 years.

QUESTIONS

1. Why does Sims Metal Management operations impact on their ability to offer competitive prices for nonferrous metals?
2. Select a recycling cycle from SIMS Metal and draw its flow chart diagram.
3. Illustrate the evolution of the global supply chain of ferrous metals in the past decade.
4. Discuss the future challenges for Sims Metal Management.

Source: This case was prepared by Xavier Pierron for in-class discussion at Coventry Business School, Coventry, UK.
Annual Reports (2011) Sims Metal Management, available: http://www.simsmm.com/news--reports/annual-reports/, last accessed October 30, 2011. *UK Sims Metal Management* (2011) Sims Metal Management, available: http://uk.simsmm.com/, last accessed October 30, 2011.

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Sharp/Getty Images News/Getty Images

6

PLANNING CAPACITY

Lines of LCD TVs at the 150 billion yen plant of Sharp electronics company in Kameyama, Mie Prefecture, Central Japan. These top notch LCD TV sets, equipped with 6.22 mega-dot LCD panel on its very thin 8.65 cm in thickness body, are designed to meet the surging demand for bigger but slimmer TVs.

Sharp Corporation

Sharp Corporation, founded in 1912 in Japan, is a global manufacturer and distributor of consumer and information products such as LCD TVs and projectors, DVD recorders, mobile communication handsets, point-of-sale systems, home appliances such as refrigerators and microwave ovens, and electronic components such as flash memories, LCD panels, and optical sensors, among others. It has manufacturing, sales, and R&D presence in 25 different countries and regions of the world. For production of its large size LCDs needed in television sets, Sharp increased the capacity of its Kameyama No. 2 plant in Japan from 15,000 LCD sheets per month in August 2006 to 30,000 LCD sheets per month during the second phase in January 2007. Capacity enhancements in the third phase (July 2007) were 60,000 LCD sheets per month. The world's largest LCD TV screen at that time, measuring 108 inch, was made at the Kameyama plant No. 2 from eight generation glass substrates.

Is such a rapid expansion of long-term capacity a good idea? Opinions vary. Even though Sharp had thrived in recent years due to its LCD business, the decision to expand was made because production capacity had reached its limit and Sharp was unable to meet its market demand. As a result, Sharp had been overtaken in global LCD television sales, and its share in the second quarter of 2006 was only 10.8 percent, fourth behind Sony, Samsung, and Philips Electronics. However, the competition did not stay idle even as Sharp quickly expanded its capacity for LCD sheets. A joint company of Sony and Samsung planned for opening a new plant in autumn of 2007 that would have the same capacity as

Sharp's Kameyama plant. Critics fear that if all three major players continue to add capacity faster than the rate at which the market can grow, it will lead to a mature industry with too much capacity and too little profitability.

Meanwhile, Sharp is betting that costs for flat panel LCDs will decline rapidly due to mass production, a strategy that it is also using to increase the capacity of thin-film type solar cells at the Katsuragi plant that can support electricity generation output from 15 MW per year to 160 MW per year. Sharp is also hoping that large capacity investments in LCD sheets will help enhance its leadership position in the LCD industry well into the future, even though it may come at the expense of lowered profitability for the entire industry.

Source: William Trent, "LCD Producers Continue to Add Too Much Capacity As They Battle for Market Share," **http://seekingalpha.com/article/19038**-lcd-producers-continue-to-add-too-much-capacity-as-they-battle-for-market-share, October 2006; **http://www.sharp-world.com/corporate/info/ci/g_organization/index.html**, and **http://www.sharp-world.com/corporate/ir/event/policy_meeting/pdf/shar080108e_1.pdf**, May 31, 2011.

LEARNING GOALS *After reading this chapter, you should be able to:*

1. Define capacity and utilization.
2. Describe economies and diseconomies of scale.
3. Identify different capacity timing and sizing strategies.
4. Identify a systematic approach to capacity planning.
5. Describe how waiting-line models, simulation, and decision trees assist capacity decisions.

Competing with Operations

Using Operations to Compete
Managing Effective Projects

Designing and Managing Processes

Developing a Process Strategy
Analyzing Processes
Managing Quality
Planning Capacity
Managing Process Constraints
Designing Lean Systems

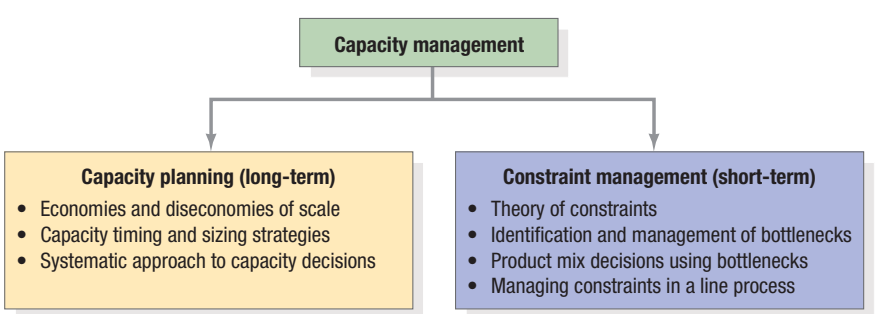
Designing and Managing Supply Chains

Managing Inventories
Designing Effective Supply Chains
Locating Facilities
Integrating the Supply Chain
Managing Sustainable Supply Chains
Forecasting Demand
Planning and Scheduling Operations
Planning Sufficient Resources

Capacity is the maximum rate of output of a process or a system. Managers are responsible for ensuring that the firm has the capacity to meet current and future demand. Otherwise, the organization will miss out on opportunities for growth and profits. Making adjustments to decrease capacity, or to overcome capacity shortfalls as Sharp Corporation did when trailing its nearest competitors for LCD TVs, is therefore an important part of the job. Acquisition of new capacity requires extensive planning, and often involves significant expenditure of resources and time. Bringing new capacity online can take several years, for instance in the semiconductor industry or in the construction of new nuclear power plants.

Capacity decisions related to a process need to be made in light of the role the process plays within the organization and the supply chain as a whole, because changing the capacity of a process will have an impact on other processes in the chain. Increasing or decreasing capacity by itself is not as important as ensuring that the entire supply chain, from order entry to delivery, is designed for effectiveness. Capacity decisions must be made in light of several long-term issues such as the firm's economies and diseconomies of scale, capacity cushions, timing and sizing strategies, and trade-offs between customer service and capacity utilization.

This chapter focuses on how managers can best revise capacity levels and best determine when to add or reduce capacity for the long term. The type of capacity decisions differ for different time horizons. Both long-term as well as short-term issues associated with planning capacity and managing constraints are important, and must be understood in conjunction with one another. While we deal here with the long-term decisions shown in the capacity management framework, short-term decisions centered on making the most of existing capacity by managing constraints are more fully explored in Chapter 7, "Managing Process Constraints."



Planning Capacity across the Organization

capacity
The maximum rate of output of a process or a system.

Capacity decisions have implications for different functional areas throughout the organization. Accounting needs to provide the cost information needed to evaluate capacity expansion decisions. Finance performs the financial analysis of proposed capacity expansion investments and raises funds to support them. Marketing provides demand forecasts needed to identify capacity gaps. Management information systems designs the electronic infrastructure that is needed to make data such as cost information, financial performance measures, demand forecasts, and work standards available to those needing it to analyze capacity options. Operations is involved in the selection of capacity strategies that can be implemented to effectively meet future demand. Purchasing facilitates acquisition of outside capacity from suppliers. Finally, human resources focuses on hiring and training employees needed to support internal capacity plans. So all departments in a firm get involved with and are affected by long-term capacity planning decisions.

Planning Long-Term Capacity

Long-term capacity plans deal with investments in new facilities and equipment at the organizational level, and require top management participation and approval because they are not easily reversed. These plans cover at least two years into the future, but construction lead times can sometimes be longer and result in longer planning time horizons.

As already seen in our opening vignette, long-term capacity planning is central to the success of an organization. Too much capacity can be as agonizing as too little. Often entire industries can fluctuate over time between too much and too little capacity, as evidenced in the airline and cruise ship industry over the past 20 years. When choosing a capacity strategy, managers must consider questions such as the following: How much of a cushion is needed to handle variable, or uncertain, demand? Should we expand capacity ahead of demand, or wait until demand is more certain? Even before these questions can be answered, a manager needs to be able to measure a process's capacity. So a systematic approach is needed to answer these and similar questions and to develop a capacity strategy appropriate for each situation.

Measures of Capacity and Utilization

No single capacity measure is best for all situations. A retailer measures capacity as annual sales dollars generated per square foot, whereas an airline measures capacity as available seat-miles (ASMs) per month. A theater measures capacity as number of seats, while a job shop measures capacity as number of machine hours. In general, capacity can be expressed in one of two ways: in terms of output measures or input measures.

Output Measures of Capacity *Output measures* of capacity are best utilized when applied to individual processes within the firm, or when the firm provides a relatively small number of standardized services and products. High-volume processes, such as those in a car manufacturing plant, are a good example. In this case, capacity would be measured in terms of the number of cars produced per day. However, many processes produce more than one service or product. As the amount of customization and variety in the product mix increases, output-based capacity measures become less useful. Then input measures of capacity become the usual choice for measuring capacity.



Sonia A. Tellez

For the first time since 1992, Intel built a new 2.5 billion chip fabrication plant in the high-tech Jinzhou New District north of the port city of Dalian. With more than 1700 employees, it covers an area of 163,000 square meters, which is roughly the same as 23 soccer fields. Started in September 2007, the plant started production in October 2010 of 300-millimeter integrated wafers and chip sets for laptop computers, high-performance desktop PCs and powerful servers.

Input Measures of Capacity *Input measures* are generally used for low-volume, flexible processes, such as those associated with a custom furniture maker. In this case, the furniture maker might measure capacity in terms of inputs such as number of workstations or number of workers. The problem with input measures is that demand is invariably expressed as an output rate. If the furniture maker wants to keep up with demand, he or she must convert the business's annual demand for furniture into labor hours and number of employees required to fulfill those hours. We will explain precisely how this input–output conversion is done later in the chapter.

utilization
The degree to which equipment, space, or the workforce is currently being used, and is measured as the ratio of average output rate to maximum capacity (expressed as a percent).

Utilization **Utilization** is the degree to which a resource such as equipment, space, or the workforce is currently being used, and is measured as the ratio of average output rate to maximum capacity (expressed as a percent). The average output rate and the capacity must be measured in the same terms—that is, time, customers, units, or dollars. The utilization rate indicates the need for adding extra capacity or eliminating unneeded capacity.

$$\text{Utilization} = \frac{\text{Average output rate}}{\text{Maximum capacity}} \times 100\%$$

Here, we refer to maximum capacity as the greatest level of output that a process can reasonably sustain for a longer period, using realistic employee work schedules and the equipment currently in place. In some processes, this capacity level implies a one-shift operation; in others, it implies a three-shift operation. A process can be operated above its capacity level using marginal methods of production, such as overtime, extra shifts, temporarily reduced maintenance activities, overstaffing, and subcontracting. Although they help with temporary peaks, these options cannot be sustained for long. For instance, being able to handle 40 customers for a one-week peak is quite different from sustaining it for six months. Employees do not want to work excessive overtime for extended periods, so quality drops. In addition, the costs associated with overtime drive up the firm's costs. So operating processes close to (or even temporarily above) their maximum capacity can result in low customer satisfaction, minimal profits, and even losing money despite high sales levels. Such was the case with U.S. aircraft manufacturers in the late 1980s, which culminated in Boeing acquiring McDonnell Douglas in 1997 in order to shore up skyrocketing costs and plummeting profits.

Economies of Scale

economies of scale
A concept that states that the average unit cost of a service or good can be reduced by increasing its output rate.

Deciding on the best level of capacity involves consideration for the efficiency of the operations. A concept known as **economies of scale** states that the average unit cost of a service or good can be reduced by increasing its output rate. Four principal reasons explain why economies of scale can drive costs down when output increases: (1) Fixed costs are spread over more units; (2) construction costs are reduced; (3) costs of purchased materials are cut; and (4) process advantages are found.

Spreading Fixed Costs In the short term, certain costs do not vary with changes in the output rate. These fixed costs include heating costs, debt service, and managers' salaries. The depreciation of plant and equipment already owned is also a fixed cost in the accounting sense. When the average output rate—and, therefore, the facility's utilization rate—increases, the average unit cost drops because fixed costs are spread over more units.

Reducing Construction Costs Certain activities and expenses are required to build small and large facilities alike: building permits, architects' fees, and rental of building equipment. Doubling the size of the facility usually does not double construction costs.

Cutting Costs of Purchased Materials Higher volumes can reduce the costs of purchased materials and services. They give the purchaser a better bargaining position and the opportunity to take advantage of quantity discounts. Retailers such as Walmart reap significant economies of scale because their national and international stores buy and sell huge volumes of each item.

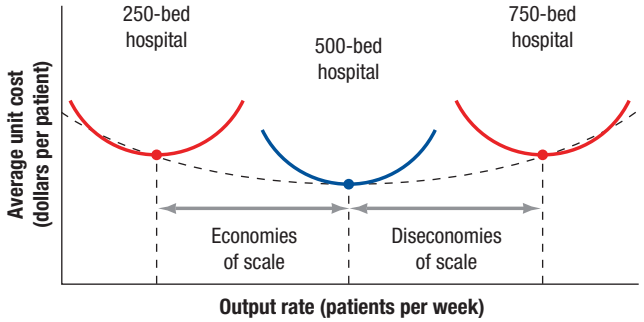
Finding Process Advantages High-volume production provides many opportunities for cost reduction. At a higher output rate, the process shifts toward a line process, with resources dedicated to individual products. Firms may be able to justify the expense of more efficient technology or more specialized equipment. The benefits from dedicating resources to individual services or products may include speeding up the learning effect, lowering inventory, improving process and job designs, and reducing the number of changeovers.

Diseconomies of Scale

diseconomies of scale
Occurs when the average cost per unit increases as the facility's size increases.

Bigger is not always better, however. At some point, a facility can become so large that **diseconomies of scale** set in; that is, the average cost per unit increases as the facility's size increases. The reason is that excessive size can bring complexity, loss of focus, and inefficiencies that raise the average unit cost of a service or product. Too many layers of employees and bureaucracy can cause management to lose touch with employees and customers. A less agile organization loses the flexibility needed to respond to changing demand. Many large companies become so involved in analysis and planning that they innovate less and avoid risks. The result is that small companies outperform corporate giants in numerous industries.

Figure 6.1 illustrates the transition from economies of scale to diseconomies of scale. The 500-bed hospital shows economies of scale because the average unit cost at its *best operating level*, represented by the blue dot, is less than that of the 250-bed hospital. However, assuming that sufficient demand exists, further expansion to a 750-bed hospital leads to higher average unit costs and diseconomies of scale. One reason the 500-bed hospital enjoys greater economies of scale than the 250-bed hospital is that the cost of building and equipping it is less than twice the cost for the smaller hospital. The 750-bed facility would enjoy similar savings. Its higher average unit costs can be explained only by diseconomies of scale, which outweigh the savings realized in construction costs.



▲ **FIGURE 6.1**
Economies and Diseconomies of Scale

Figure 6.1 does not mean that the optimal size for all hospitals is 500 beds. Optimal size depends on the number of patients per week to be served. On the one hand, a hospital serving a small community could have lower costs by choosing a 250-bed capacity rather than the 500-bed capacity. On the other hand, a large community might be served more efficiently by two 500-bed hospitals than by one 1,000-bed facility if diseconomies of scale exist at the bigger size.

Capacity Timing and Sizing Strategies

Operations managers must examine three dimensions of capacity strategy before making capacity decisions: (1) sizing capacity cushions, (2) timing and sizing expansion, and (3) linking process capacity and other operating decisions.

Sizing Capacity Cushions

Average utilization rates for any resource should not get too close to 100 percent over the long term, though it may occur for some processes from time to time in the short run. If the demand keeps increasing over time, then long-term capacity must be increased as well to provide some buffer against uncertainties. When average utilization rates approach 100 percent, it is usually a signal to increase capacity or decrease order acceptance to avoid declining productivity. The **capacity cushion** is the amount of reserve capacity a process uses to handle sudden increases in demand or temporary losses of production capacity; it measures the amount by which the average utilization (in terms of total capacity) falls below 100 percent. Specifically,

capacity cushion
The amount of reserve capacity a process uses to handle sudden increases in demand or temporary losses of production capacity; it measures the amount by which the average utilization (in terms of total capacity) falls below 100 percent.

$$\text{Capacity cushion} = 100\ (\%) - \text{Average Utilization rate}\ (\%)$$

The appropriate size of the cushion varies by industry. In the capital-intensive paper industry, where machines can cost hundreds of millions of dollars each, cushions well under 10 percent are preferred. The less capital-intensive hotel industry breaks even with a 60 to 70 percent utilization (40 to 30 percent cushion), and begins to suffer customer-service problems when the cushion drops to 20 percent. The more capital-intensive cruise ship industry prefers cushions as small as 5 percent. Large cushions are particularly vital for front-office processes where customers expect fast service times.

Businesses find large cushions appropriate when demand varies. In certain service industries (the grocery industry, for example), demand on some days of the week is predictably higher than on other days, and even hour-to-hour changes are typical. Long customer waiting times are not acceptable because customers grow impatient if they have to wait in a supermarket checkout line for more than a few minutes. Prompt customer service requires supermarkets to maintain a capacity cushion large enough to handle peak demand. Large cushions also are necessary when future demand is uncertain, particularly if resource flexibility is low. Simulation and waiting-line analysis (see Supplement B, "Waiting Line Models") can help managers better anticipate the relationship between capacity cushion and customer service.

Another type of demand uncertainty occurs with a changing product mix. Though total demand measured in monetary terms might remain stable, the load can shift unpredictably from one workstation to another as the product mix changes. Supply uncertainty tied to delivery of purchased materials also makes large capacity cushions helpful. Capacity often comes in large increments because a complete machine has to be purchased even if only a fraction of its available capacity is needed, which in turn creates a large cushion. Firms also need to build in excess capacity to allow for employee absenteeism, vacations, holidays, and any other delays. If a firm is experiencing high overtime costs and frequently needs to rely on subcontractors, it perhaps needs to increase its capacity cushions.



Jeff Kowalsky/Bloomberg/Getty Images

Dawn Bacon works on a General Motors Co. 2012 Opel Ampera, GM's European version of the Volt, at Detroit-Hamtramck Assembly Plant in Detroit, Michigan, U.S. GM will invest $69 million and add 2,500 jobs to start making two new models at the Detroit plant that builds the Chevrolet Volt plug-in hybrid as the automaker boosts U.S. production.

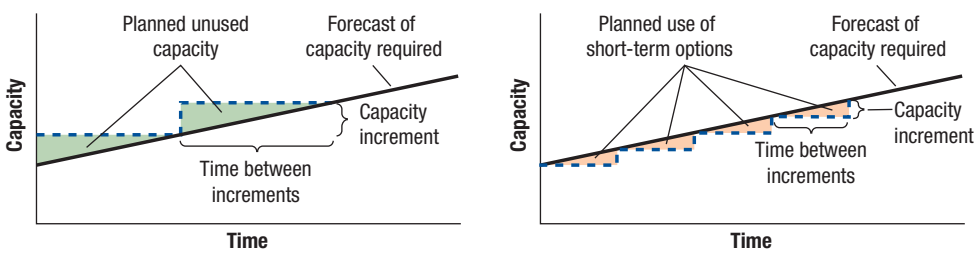
The argument in favor of small cushions is simple: Unused capacity costs money. For capital-intensive firms, minimizing the capacity cushion is vital. Studies indicate that businesses with high capital intensity achieve a low return on investment when the capacity cushion is high. This strong correlation does not exist for labor-intensive firms, however. Their return on investment is about the same because the lower investment in equipment makes high utilization less critical. Small cushions have other advantages. By implementing a small cushion, a company can sometimes uncover inefficiencies that were difficult to detect when cushions were larger. These inefficiencies might include employee absenteeism or unreliable suppliers. Once managers and workers identify such problems, they often can find ways to correct them.

Timing and Sizing Expansion

The second issue of capacity strategy concerns when to adjust capacity levels and by how much. At times, capacity expansion can be done in response to changing market trends. General Motors decided to increase production capacity of the four-seat series hybrid car Chevrolet Volt from 30,000 units to 45,000 units in 2012 because of strong public interest. While we deal with this issue from the perspective of capacity expansion in greater detail here, it must be noted that firms may not always be looking to expand capacity, but at times may be forced to retrench as evidenced by the situation in the airlines industry, where all major airlines have consolidated routes and reduced the total number of flights in the face of increasing oil costs. Some of this consolidation has been achieved through mergers like United Airlines and Continental to create the world's largest airline company, as well as between Delta and Northwest Airlines.

Figure 6.2 illustrates two extreme strategies for expanding capacity: the *expansionist strategy*, which involves large, infrequent jumps in capacity, and the *wait-and-see strategy*, which involves smaller, more frequent jumps.

FIGURE 6.2 ▶
Two Capacity Strategies



(a) Expansionist strategy

(b) Wait-and-see strategy

The timing and sizing of expansion are related; that is, if demand is increasing and the time between increments increases, the size of the increments must also increase. The expansionist strategy, which stays ahead of demand, minimizes the chance of sales lost to insufficient capacity. The wait-and-see strategy lags behind demand. To meet any shortfalls, it relies on short-term options, such as use of overtime, temporary workers, subcontractors, stockouts, and the postponement of preventive maintenance on equipment.

Several factors favor the expansionist strategy. Expansion can result in economies of scale and a faster rate of learning, thus helping a firm reduce its costs and compete on price. This strategy might increase the firm's market share or act as a form of preemptive marketing. By making a large capacity expansion or announcing that one is imminent, the firm can preempt the expansion of other firms. These other firms must sacrifice some of their market share or risk burdening the industry with overcapacity. To be successful, however, the preempting firm must have the credibility to convince the competition that it will carry out its plans—and must signal its plans before the competition can act. Managerial Practice 6.1 illustrates the use of expansionist strategy in the ethanol industry.

MANAGERIAL PRACTICE 6.1 Expansionist Capacity Strategy in the Ethanol Industry

A search for renewable energy and alternative sources of fuel has made ethanol production a big growth industry in the United States. The state of Nebraska alone has capital investments of more than $5 billion in 24 ethanol production plants, which together produce more than 2 billion gallons of ethanol each year. While Nebraska serves the ethanol markets in western United States, midwestern states like Iowa have the largest invested capacity. Many of these plants have gone into operation since 2007.

But troubling signs may be on the horizon. Companies and farm cooperatives have built so many distilleries so quickly that the ethanol market is suddenly plagued by a glut and oversupply is putting downward pressure on price. At the same time, scarcity and price of raw materials needed to feed ethanol plants is becoming an issue. One bushel of corn produces approximately 2.6 gallons of ethanol, and the ethanol plants in the state of Nebraska alone require over 769 million bushels of grain in the process every year. As a result, the price of corn has risen steadily from under $2 per bushel in 2006 to $7.76 per bushel in April 2011. The price per bushel doubled in one year alone between 2010 and 2011, though it is expected to come down in the future as more corn is planted and the corn reserves rise due to lesser exports. These price increases have further eroded profitability. Despite the steep increase in prices of crude oil, demand for ethanol has not increased proportionately because it is less efficient and constitutes 10 percent or less in most gasoline blends.

Capacity was also expanded without taking into consideration the transportation needs of ethanol. Unlike gasoline, traditional pipelines cannot be used due to the fact that ethanol is corrosive and contains traces of water. So it must be transported by specialized ethanol rail cars, trucks, and barges. Capacity of such a transportation network has not kept up with the surge in surplus ethanol production. Consequently, demand for ethanol along coastal regions of the country cannot be met quickly and efficiently from the heartland locations of Corn Belt ethanol plants.



Andy Kropa/Redux

The Lincolnway Energy plant, a midsize distillery in Nevada, Iowa, converts corn into ethanol in a largely automated process. Lincolnway Energy was once virtually alone in the region. Today, though, competing distilleries are operating and pouring even more ethanol onto the market.

While the government subsidies may keep the output of ethanol growing, experts believe that the expansion of the industry has been poorly managed. Some companies are modifying future plans and canceling new plant construction in fear of consolidation and going out of business.

Source: Clifford Krauss, "Ethanol's Boom Stalling as Glut Depresses Price," *New York Times* (September 30, 2007); Christopher Leonard, "Corn Reserves Expected to Rise, Easing Food Prices," *The Associated Press,* May 12, 2011; Nebraska Ethanol Board, "Ethanol Plants in Nebraska," **http://www.ne-ethanol.org/industry/ethplants.htm** (May 31, 2011).

The conservative wait-and-see strategy is to expand in smaller increments, such as by renovating existing facilities rather than building new ones. Because the wait-and-see strategy follows demand, it reduces the risks of overexpansion based on overly optimistic demand forecasts, obsolete technology, or inaccurate assumptions regarding the competition.

However, this strategy has its own risks, such as being preempted by a competitor or being unable to respond if demand is unexpectedly high. Critics claim the wait-and-see strategy is a short-term strategy typical of some U.S. management styles. Managers on the fast track to corporate advancement tend to take fewer risks. They earn promotions by avoiding the big mistakes and maximizing short-term profits and return on investment. The wait-and-see strategy fits this short-term outlook but can erode market share over the long run.

Management may choose one of these two strategies or one of the many between these extremes. With strategies in the more moderate middle, firms can expand more frequently (on a smaller scale) than they can with the expansionist strategy without lagging behind demand as with the wait-and-see strategy. An intermediate strategy could be to *follow the leader,* expanding when others do. If others are right, so are you, and nobody gains a competitive advantage. If others make a mistake and over expand, so do you, but everyone shares in the agony of overcapacity. Such a situation was noted for the airlines industry, and may yet occur in the LCD industry due to large capacity expansions by Sharp Corporation, Sony, and Samsung as described in the opening vignette.

Linking Capacity and Other Decisions

Capacity decisions should be closely linked to processes and supply chains throughout the organization. When managers make decisions about designing processes, determining degree of resource flexibility and inventory, and locating facilities, they must consider its impact on capacity

cushions. Capacity cushions in the long run buffer the organization against uncertainty, as do resource flexibility, inventory, and longer customer lead times. If a change is made in any one decision area, the capacity cushion may also need to be changed to compensate. For example, capacity cushions for a process can be lowered if less emphasis is placed on fast deliveries (*competitive priorities*), yield losses (*quality*) drop, or if investment in capital-intensive equipment increases or worker flexibility increases (*process design*). Capacity cushions can also be lowered if the company is willing to smooth the output rate by raising prices when inventory is low and decreasing prices when it is high.

A Systematic Approach to Long-Term Capacity Decisions

Long-term decisions for capacity would typically include whether to add a new plant or warehouse or to reduce the number of existing ones, how many workstations a given department should have, or how many workers are needed to staff a given process. Some of these decisions can take years to become operational. Hence, a systematic approach is needed to plan for long-term capacity decisions.

Although each situation is somewhat different, a four-step procedure generally can help managers make sound capacity decisions. (In describing this procedure, we assume that management already performed the preliminary steps of determining the process's existing capacity and assessing whether its current capacity cushion is appropriate.)

1. Estimate future capacity requirements.
2. Identify gaps by comparing requirements with available capacity.
3. Develop alternative plans for reducing the gaps.
4. Evaluate each alternative, both qualitatively and quantitatively, and make a final choice.

Step 1: Estimate Capacity Requirements

capacity requirement
What a process's capacity should be for some future time period to meet the demand of customers (external or internal), given the firm's desired capacity cushion.

planning horizon
The set of consecutive time periods considered for planning purposes.

A process's **capacity requirement** is what its capacity should be for some future time period to meet the demand of the firm's customers (external or internal), given the firm's desired capacity cushion. Larger requirements are practical for processes or workstations that could potentially be bottlenecks in the future, and management may even plan for larger cushions than normal.

Capacity requirements can be expressed in one of two ways: with an output measure or with an input measure. Either way, the foundation for the estimate is forecasts of demand, productivity, competition, and technological change. These forecasts normally need to be made for several time periods in a **planning horizon**, which is the set of consecutive time periods considered for planning purposes. Long-term capacity plans need to consider more of the future (perhaps, a whole decade) than do short-term plans. Unfortunately, the further ahead you look, the more chance you have of making an inaccurate forecast.

Using Output Measures The simplest way to express capacity requirements is as an output rate. As discussed earlier, output measures are appropriate for high-volume processes with little product variety or process divergence. Here, demand forecasts for future years are used as a basis for extrapolating capacity requirements into the future. If demand is expected to double in the next five years, then the capacity requirements also double. For example, if a process's current demand is 50 customers per day, then the demand in five years would be 100 customers per day. If the desired capacity cushion is 20 percent, management should plan for enough capacity to serve $[100/(1 - 0.2)] = 125$ customers in five years.

Using Input Measures Output measures may be insufficient in the following situations:

- Product variety and process divergence is high.
- The product or service mix is changing.
- Productivity rates are expected to change.
- Significant learning effects are expected.

In such cases, it is more appropriate to calculate capacity requirements using an input measure, such as the number of employees, machines, computers, or trucks. Using an input measure for the capacity requirement brings together demand forecasts, process time estimates, and the

desired capacity cushion. When just one service or product is processed at an operation and the time period is a particular year, the capacity requirement, M, is

$$\text{Capacity requirement} = \frac{\text{Processing hours required for year's demand}}{\text{Hours available from a single capacity unit (such as an employee or machine) per year, after deducting desired cushion}}$$

$$M = \frac{Dp}{N[1 - (C/100)]}$$

where

D = demand forecast for the year (number of customers served or units produced)

p = processing time (in hours per customer served or unit produced)

N = total number of hours per year during which the process operates

C = desired capacity cushion (expressed as a percent)

M is the number of input units required and should be calculated for each year in the time horizon. The processing time, p, depends on the process and methods selected to do the work. The denominator is the total number of hours, N, available for the year from one unit of capacity (an employee or machine), multiplied by a proportion that accounts for the desired capacity cushion, C. The proportion is simply $1.0 - C/100$, where C is converted from a percent to a proportion by dividing by 100. For example, a 20 percent capacity cushion means that $1.0 - C/100 = 0.80$.

Setups may be involved if multiple products are being manufactured. **Setup time** is the time required to change a process or an operation from making one service or product to making another. The total setup time is found by dividing the number of units forecast per year, D, by the number of units made in each lot, Q, (number of units processed between setups), which gives the number of setups per year, and then multiplying by the time per setup, s. For example, if the annual demand is 1,200 units and the average lot size is 100, there are $1{,}200/100 = 12$ setups per year. Accounting for both processing and setup times for multiple services (products), we get

setup time

The time required to change a process or an operation from making one service or product to making another.

$$\text{Capacity requirement} = \frac{\text{Processing and setup hours required for year's demand, summed over all services or products}}{\text{Hours available from a single capacity unit per year, after deducting desired cushion}}$$

$$M = \frac{[Dp + (D/Q)s]_{\text{product 1}} + [Dp + (D/Q)s]_{\text{product 2}} + \cdots + [Dp + (D/Q)s]_{\text{product } n}}{N\,[1-(C/100)]}$$

where

Q = number of units in each lot

s = setup time (in hours) per lot

What to do when M is not an integer depends on the situation. For example, it is impossible to buy a fractional machine. In this case, round up the fractional part, unless it is cost efficient to use short-term options, such as overtime or stockouts, to cover any shortfalls. If, instead, the capacity unit is the number of employees at a process, a value of 23.6 may be achieved using just 23 employees and a modest use of overtime (equivalent to having 60 percent of another full-time person). Here, the fractional value should be retained as useful information.

EXAMPLE 6.1 Estimating Capacity Requirements When Using Input Measures

A copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run, collate, and bind each copy depends on, among other factors, the number of pages. The center operates 250 days per year, with one 8-hour shift. Management believes that a capacity cushion of 15 percent (beyond the allowance built into time standards) is best. It currently has three copy machines. Based on the following table of information, determine how many machines are needed at the copy center.

Item	Client X	Client Y
Annual demand forecast (copies)	2,000	6,000
Standard processing time (hour/copy)	0.5	0.7
Average lot size (copies per report)	20	30
Standard setup time (hours)	0.25	0.40

SOLUTION

$$M = \frac{[Dp + (D/Q)s]_{\text{product 1}} + [Dp + (D/Q)s]_{\text{product 2}} + \cdots + [Dp + (D/Q)s]_{\text{product } n}}{N\,[1-(C/100)]}$$

$$= \frac{[2{,}000(0.5) + (2{,}000/20)(0.25)]_{\text{client X}} + [6{,}000(0.7) + (6{,}000/30)(0.40)]_{\text{client Y}}}{[(250\text{ day/year})(1\text{ shift/day})(8\text{ hours/day})][1.0 - (15/100)]}$$

$$= \frac{5{,}305}{1{,}700} = 3.12$$

Rounding up to the next integer gives a requirement of four machines.

DECISION POINT

The copy center's capacity is being stretched and no longer has the desired 15 percent capacity cushion with the existing three machines. Not wanting customer service to suffer, management decided to use overtime as a short-term solution to handle past-due orders. If demand continues at the current level or grows, it will acquire a fourth machine.

Step 2: Identify Gaps

capacity gap
Positive or negative difference between projected demand and current capacity.

A **capacity gap** is any difference (positive or negative) between projected capacity requirements (M) and current capacity. Complications arise when multiple operations and several resource inputs are involved. Expanding the capacity of some operations may increase overall capacity. However, as we will learn later in Chapter 7, "Managing Process Constraints," if one operation is more constrained than others, total process capacity can be expanded only if the capacity of the constrained operation is expanded.

Step 3: Develop Alternatives

base case
The act of doing nothing and losing orders from any demand that exceeds current capacity, or incur costs because capacity is too large.

The next step is to develop alternative plans to cope with projected gaps. One alternative, called the **base case**, is to do nothing and simply lose orders from any demand that exceeds current capacity or incur costs because capacity is too large. Other alternatives if expected demand exceeds current capacity are various timing and sizing options for adding new capacity, including the expansionist and wait-and-see strategies illustrated in Figure 6.2. Additional possibilities include expanding at a different location and using short-term options, such as overtime, temporary workers, and subcontracting. Alternatives for reducing capacity include the closing of plants or warehouses, laying off employees, or reducing the days or hours of operation.

Step 4: Evaluate the Alternatives

In this final step, the manager evaluates each alternative, both qualitatively and quantitatively.

Qualitative Concerns Qualitatively, the manager looks at how each alternative fits the overall capacity strategy and other aspects of the business not covered by the financial analysis. Of particular concern might be uncertainties about demand, competitive reaction, technological change, and cost estimates. Some of these factors cannot be quantified and must be assessed on the basis of judgment and experience. Others can be quantified, and the manager can analyze each alternative by using different assumptions about the future. One set of assumptions could represent a worst case, in which demand is less, competition is greater, and construction costs are higher than expected. Another set of assumptions could represent the most optimistic view of the future. This type of "what-if" analysis allows the manager to get an idea of each alternative's implications before making a final choice.

Qualitative factors would tend to dominate when a business is trying to enter new markets or change the focus of its business strategy. For instance, Dell is planning a $1 billion investment in cloud computing (on-demand provision of computational resources for data and software through computer networks rather than local computers) and virtualization (creating a virtual rather than an actual version of an operating system or a storage device) in the 2011 fiscal year. Little hard data may be available to guide the exact size and timing of the significant expansion of data center capacity that must be undertaken to support this diversification strategy, which will also include a deeper focus on sales training and expertise.



Yan Sheng - CNImaging/Newscom

Dell's Green Energy-saving and Environment-protecting Data Center will monitor the energy consumption index of 54 government offices in Beijing, China

Quantitative Concerns Quantitatively, the manager estimates the change in cash flows for each alternative over the forecast time horizon compared to the base case. **Cash flow** is the difference between the flows of funds into and out of an organization over a period of time, including revenues, costs, and changes in assets and liabilities. The manager is concerned here only with calculating the cash flows attributable to the project.

cash flow

The difference between the flows of funds into and out of an organization over a period of time, including revenues, costs, and changes in assets and liabilities.

EXAMPLE 6.2 Evaluating the Alternatives

Grandmother's Chicken Restaurant is experiencing a boom in business. The owner expects to serve 80,000 meals this year. Although the kitchen is operating at 100 percent capacity, the dining room can handle 105,000 diners per year. Forecasted demand for the next 5 years is 90,000 meals for next year, followed by a 10,000-meal increase in each of the succeeding years. One alternative is to expand both the kitchen and the dining room now, bringing their capacities up to 130,000 meals per year. The initial investment would be $200,000, made at the end of this year (year 0). The average meal is priced at $10, and the before-tax profit margin is 20 percent. The 20 percent figure was arrived at by determining that, for each $10 meal, $8 covers variable costs and the remaining $2 goes to pretax profit.

What are the pretax cash flows from this project for the next 5 years compared to those of the base case of doing nothing?

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Tutor 6.2 in MyOMLab provides a new example to practice projecting cash flows for capacity decisions.

SOLUTION

Recall that the base case of doing nothing results in losing all potential sales beyond 80,000 meals. With the new capacity, the cash flow would equal the extra meals served by having a 130,000-meal capacity, multiplied by a profit of $2 per meal. In year 0, the only cash flow is −$200,000 for the initial investment. In year 1, the 90,000-meal demand will be completely satisfied by the expanded capacity, so the incremental cash flow is (90,000 − 80,000) ($2) = $20,000. For subsequent years, the figures are as follows:

Year 2: Demand = 100,000; Cash flow = (100,000 − 80,000)$2 = $40,000

Year 3: Demand = 110,000; Cash flow = (110,000 − 80,000)$2 = $60,000

Year 4: Demand = 120,000; Cash flow = (120,000 − 80,000)$2 = $80,000

Year 5: Demand = 130,000; Cash flow = (130,000 − 80,000)$2 = $100,000

If the new capacity were smaller than the expected demand in any year, we would subtract the base case capacity from the new capacity (rather than the demand). The owner should account for the time value of money, applying such techniques as the net present value or internal rate of return methods (see MyOMLab Supplement F, "Financial Analysis"). For instance, the net present value (NPV) of this project at a discount rate of 10 percent is calculated here, and equals $13,051.76.

MyOMLab

$$\text{NPV} = -200{,}000 + [(20{,}000/1.1)] + [40{,}000/(1.1)^2] + [60{,}000/(1.1)^3] + [80{,}000/(1.1)^4] + [100{,}000/(1.1)^5]$$

$$= -\$200{,}000 + \$18{,}181.82 + \$33{,}057.85 + \$45{,}078.89 + \$54{,}641.07 + \$62{,}092.13$$

$$= \$13{,}051.76$$

DECISION POINT

Before deciding on this capacity alternative, the owner should also examine the qualitative concerns, such as future location of competitors. In addition, the homey atmosphere of the restaurant may be lost with expansion. Furthermore, other alternatives should be considered (see Solved Problem 2).

Tools for Capacity Planning

Capacity planning requires demand forecasts for an extended period of time. Unfortunately, forecast accuracy declines as the forecasting horizon lengthens. In addition, anticipating what competitors will do increases the uncertainty of demand forecasts. Demand during any period of time may not be evenly distributed; peaks and valleys of demand may (and often do) occur within the time period. These realities necessitate the use of capacity cushions. In this section, we introduce three tools that deal more formally with demand uncertainty and variability: (1) waiting-line models, (2) simulation, and (3) decision trees. Waiting-line models and simulation account for the random, independent behavior of many customers, in terms of both their time of arrival and their processing needs. Decision trees allow anticipation of events, such as competitors' actions, which requires a sequence of decisions regarding capacities.

Waiting-Line Models

Waiting-line models often are useful in capacity planning, such as selecting an appropriate capacity cushion for a high customer-contact process. Waiting lines tend to develop in front of a work center, such as an airport ticket counter, a machine center, or a central computer. The reason is that the arrival time between jobs or customers varies, and the processing time may vary from one customer to the next. Waiting-line models use probability distributions to provide estimates of average customer wait time, average length of waiting lines, and utilization of the work center. Managers can use this information to choose the most cost-effective capacity, balancing customer service and the cost of adding capacity.

Supplement B, "Waiting Line Models," follows this chapter and provides a fuller treatment of these models. It introduces formulas for estimating important characteristics of a waiting line, such as average customer waiting time and average facility utilization for different facility designs. For example, a facility might be designed to have one or multiple lines at each operation and to route customers through one or multiple operations. Given the estimating capability of these formulas and cost estimates for waiting and idle time, managers can select cost-effective designs and capacity levels that also provide the desired level of customer service.

Figure 6.3 shows output from POM for Windows for waiting lines. A professor meeting students during office hours has an arrival rate of three students per hour and a service rate of six students per hour. The output shows that the capacity cushion is 50 percent (1 − average server utilization of 0.50). This result is expected because the processing rate is double the arrival rate. What might not be expected is that a typical student spends 20 minutes either in line or talking with the professor, and the probability of having two or more students at the office is 0.25. These numbers might be surprisingly high, given such a large capacity cushion.

FIGURE 6.3 ▼
POM for Windows Output for Waiting Lines during Office Hours

Waiting Lines Results

Example Solution

Parameter	Value	Parameter	Value	Minutes	Seconds
Single-Server Model		Average server utilization	.5		
Arrival rate(lambda)	3	Average number in the line(Lq)	.5		
Service rate(mu)	6	Average number in the system(L)	1		
Number of servers	1	Average time in the line(Wq)	.17	10	600
		Average time in the system(W)	.33	20	1200

Table of Probabilities

Example Solution

k	Prob (num in sys = k)	Prob (num in sys <= k)	Prob (num in sys >k)
0	.5	.5	.5
1	.25	.75	.25
2	.13	.88	.13
3	.06	.94	.06
4	.03	.97	.03
5	.02	.98	.02
6	.01	1	.01
7	.0	1	.0

Simulation

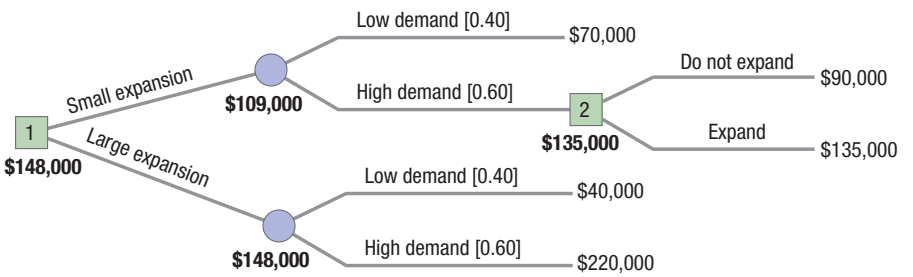
More complex waiting-line problems must be analyzed with simulation. It can identify the process's bottlenecks and appropriate capacity cushions, even for complex processes with random demand patterns and predictable surges in demand during a typical day. The SimQuick simulation package, provided in MyOMLab, allows you to build dynamic models and systems. Other simulation packages can be found with Extend, Simprocess, ProModel, and Witness.

MyOMLab

Decision Trees

A decision tree can be particularly valuable for evaluating different capacity expansion alternatives when demand is uncertain and sequential decisions are involved (see Supplement A, "Decision Making Models"). For example, the owner of Grandmother's Chicken Restaurant (see Example 6.2) may expand the restaurant now, only to discover in year 4 that demand growth is much higher than forecasted. In that case, she needs to decide whether to expand further. In terms of construction costs and downtime, expanding twice is likely to be much more expensive than building a larger facility from the outset. However, making a large expansion now, when demand growth is low, means poor facility utilization. Much depends on the demand.

Figure 6.4 shows a decision tree for this view of the problem, with new information provided. Demand growth can be either low or high, with probabilities of 0.40 and 0.60, respectively. The initial expansion in year 1 (square node 1) can either be small or large. The second decision node (square node 2), whether to expand at a later date, is reached only if the initial expansion is small and demand turns out to be high. If demand is high and if the initial expansion was small, a decision must be made about a second expansion in year 4. Payoffs for each branch of the tree are estimated. For example, if the initial expansion is large, the financial benefit is either $40,000 or $220,000, depending on whether demand is low or high. Weighting these payoffs by the probabilities yields an expected value of $148,000. This expected payoff is higher than the $109,000 payoff for the small initial expansion, so the better choice is to make a large expansion in year 1.



◀ **FIGURE 6.4**
A Decision Tree for Capacity Expansion

LEARNING GOALS IN REVIEW

1. **Define capacity and utilization.** Review the section "Measures of Capacity and Utilization," pp. 223–224, and understand why and how capacity is measured in high volume processes is different from its measurement in low volume flexible processes.
2. **Describe economies and diseconomies of scale.** See the section on "Planning Long-Term Capacity," pp. 223–225. Figure 6.1 illustrates the relationship between average unit cost and output rate, and shows different output ranges over which economies and diseconomies of scale can occur.
3. **Identify different capacity timing and sizing strategies.** The section "Capacity Timing and Sizing Strategies," pp. 225–228, and Figure 6.2 differentiates between expansionist and wait-and-see strategies. Understand the notion of capacity cushions, and how they link to other decisions in the firm.
4. **Identify a systematic approach to capacity planning.** The section "A Systematic Approach to Long-Term Capacity Decisions," pp. 228–231, shows you how capacity requirements can be estimated for both input based as well as output based measures. Focus on how different alternatives can be developed to fill the capacity gaps between requirements and current capacity.
5. **Describe how waiting-line models, simulation, and decision trees assist capacity decisions.** The section "Tools for Capacity Planning," pp. 232–233, illustrate how these different methods can be used to arrive at capacity decisions.

MyOMLab helps you develop analytical skills and assesses your progress with multiple problems on capacity utilization, capacity requirements, future pretax cash flows, with and without expansion, decisions trees, NPV, and payback periods.

MyOMLab Resources	Titles	Link to the Book
Video	*Gate Turnaround at Southwest Airlines*	Planning Long-Term Capacity; Capacity Timing and Sizing Strategies
	1st Bank Villa Italia: Waiting Lines	Supplement B; Waiting-Line Models
OM Explorer Solvers	Capacity Requirements	A Systematic Approach to Long-Term Capacity Decisions
OM Explorer Tutors	6.1 Capacity Requirements	A Systematic Approach to Long-Term Capacity Decisions; Example 6.1 (pp. 229–230)
	6.2 Projecting Cash Flows	Evaluate the Alternatives; Example 6.2 (p. 231)
Tutor Exercises	6.1 Plowton Industries	Planning Long-Term Capacity; Capacity Timing and Sizing Strategies; Solved Problem 1 (pp. 235–236)
Virtual Tours	FaucetCraft Manufacturing	Planning Long-Term Capacity; Capacity Timing and Sizing Strategies; Example 6.1 (pp. 229–230)
	Portland Bolt & Manufacturing	Planning Long-Term Capacity; Capacity Timing and Sizing Strategies; Example 6.1 (pp. 229–230)
MyOMLab Supplements	F. Financial Analysis	Evaluating the Alternatives
	H. Measuring Output Rates	MyOMLab Supplement H
	I. Learning Curve Analysis	MyOMLab Supplement I
Internet Exercise	Vintak Intex Co., Ltd.	Planning Long-Term Capacity; Capacity Timing and Sizing Strategies; Figure 6.2 (p. 226)
	Design Homes, Inc.	Capacity Timing and Sizing Strategies; Example 6.1 (pp. 229–230); Example 6.2 (p. 231)
Additional Case	Fitness Plus (B)	Example 6.1 (pp. 229–230); Example 6.2 (p. 231); Solved Problem 2 (pp. 236–237)
Key Equations		
Image Library		

Key Equations

1. Utilization, expressed as a percent:

$$\text{Utilization} = \frac{\text{Average output rate}}{\text{Maximum capacity}} \times 100\%$$

2. Capacity cushion, C, expressed as a percent:

$$C = 100\% - \text{Average Utilization rate}(\%)$$

a. Capacity requirement for one service or product:

$$M = \frac{Dp}{N[1 - (C/100)]}$$

b. Capacity requirement for multiple services or products:

$$M = \frac{[Dp + (D/Q)s]_{\text{product 1}} + [Dp + (D/Q)s]_{\text{product 2}} + \cdots + [Dp + (D/Q)s]_{\text{product } n}}{N\,[1-(C/100)]}$$

Key Terms

base case 230
capacity 222
capacity cushion 225
capacity gap 230
capacity requirement 228
cash flow 231
diseconomies of scale 224
economies of scale 224
planning horizon 228
setup time 229
utilization 224

Solved Problem 1

You have been asked to put together a capacity plan for a critical operation at the Surefoot Sandal Company. Your capacity measure is number of machines. Three products (men's, women's, and children's sandals) are manufactured. The time standards (processing and setup), lot sizes, and demand forecasts are given in the following table. The firm operates two 8-hour shifts, 5 days per week, 50 weeks per year. Experience shows that a capacity cushion of 5 percent is sufficient.

	TIME STANDARDS			
Product	**Processing (hr/pair)**	**Setup (hr/pair)**	**Lot Size (pairs/lot)**	**Demand Forecast (pairs/yr)**
Men's sandals	0.05	0.5	240	80,000
Women's sandals	0.10	2.2	180	60,000
Children's sandals	0.02	3.8	360	120,000

a. If the operation currently has two machines, what is the capacity gap?

b. How many machines are needed?

SOLUTION

a. The number of hours of operation per year, N, is $N = (2 \text{ shifts/day})(8 \text{ hours/shifts})(250 \text{ days/machine-year}) = 4{,}000 \text{ hours/machine-year}$

The number of machines required, M, is the sum of machine-hour requirements for all three products divided by the number of productive hours available for one machine:

$$M = \frac{[Dp + (D/Q)s]_{\text{men}} + [Dp + (D/Q)s]_{\text{women}} + [Dp + (D/Q)s]_{\text{children}}}{N\,[1-(C/100)]}$$

$$= \frac{\begin{array}{r}[80{,}000(0.05) + (80{,}000/240)0.5] + [60{,}000(0.10) + (60{,}000/180)2.2] \\ + [120{,}000(0.02) + (120{,}000/360)3.8]\end{array}}{4{,}000[1 - (5/100)]}$$

$$= \frac{14{,}567 \text{ hours/year}}{3{,}800 \text{ hours/machine} - \text{year}} = 3.83 \text{ or } 4 \text{ machines}$$

b. The capacity gap is 1.83 machines (3.83 − 2). Two more machines should be purchased, unless management decides to use short-term options to fill the gap.

The *Capacity Requirements* Solver in OM Explorer confirms these calculations, as Figure 6.5 shows, using only the "Expected" scenario for the demand forecasts.

FIGURE 6.5 ▶
Using the *Capacity Requirements* Solver for Solved Problem 1

Shifts/Day	2
Hours/Shift	8
Days/Week	5
Weeks/Year	50
Cushion (as %)	5%
Current capacity	2

Components: 3 — More Components / Fewer Components

Components	Processing (hr/unit)	Setup (hr/lot)	Lot Size (units/lot)	Demand Forecasts Pessimistic	Expected	Optimistic
Men's sandals	0.05	0.5	240		80,000	
Women's sandals	0.10	2.2	180		60,000	
Children's sandals	0.02	3.8	360		120,000	

Productive hours from one capacity unit for a year: 3,800

	Pessimistic Process	Pessimistic Setup	Expected Process	Expected Setup	Optimistic Process	Optimistic Setup
Men's sandals	0	0.0	4,000	166.7	0	0.0
Women's sandals	0	0.0	6,000	733.3	0	0.0
Children's sandals	0	0.0	2,400	1,266.7	0	0.0
	0	0.0	12,400	2,166.7	0	0.0
Total hours required		0.0		14,566.7		0.0
Total capacity requirements (M)		0.00		3.83		0.00
Rounded		0		4		0
Scenarios that can be met with current system/capacity:				Pessimistic, Optimistic		
If capacity increased by		0%				
Expanded current capacity		3,800				
Total capacity requirements (M)		0.00		3.83		0.00
Rounded		0		4		0
Scenarios that can be met with expanded current capacity:				Pessimistic, Optimistic		

Solved Problem 2

The base case for Grandmother's Chicken Restaurant (see Example 6.2) is to do nothing. The capacity of the kitchen in the base case is 80,000 meals per year. A capacity alternative for Grandmother's Chicken Restaurant is a two-stage expansion. This alternative expands the kitchen at the end of year 0, raising its capacity from 80,000 meals per year to that of the dining area (105,000 meals per year). If sales in year 1 and 2 live up to expectations, the capacities of both the kitchen and the dining room will be expanded at the *end* of year 3 to 130,000 meals per year. This upgraded capacity level should suffice up through year 5. The initial investment would be $80,000 at the end of year 0 and an additional investment of $170,000 at the end of year 3. The pretax profit is $2 per meal. What are the pretax cash flows for this alternative through year 5, compared with the base case?

SOLUTION

Table 6.1 shows the cash inflows and outflows. The year 3 cash flow is unusual in two respects. First, the cash inflow from sales is $50,000 rather than $60,000. The increase in sales over the base is 25,000 meals (105,000 − 10,000) instead of 30,000 meals (110,000 − 80,000) because the restaurant's capacity falls somewhat short of demand. Second, a cash outflow of $170,000 occurs at the end of year 3, when the second-stage expansion occurs. The net cash flow for year 3 is $50,000 − $170,000 = −$120,000.

For comparison purposes, the NPV of this project at a discount rate of 10 percent is calculated as follows, and equals negative $2,184.90.

$$\begin{aligned}\text{NPV} &= -80{,}000 + (20{,}000/1.1) + [40{,}000/(1.1)^2] - [120{,}000/(1.1)^3] + [80{,}000/(1.1)^4] + [100{,}000/(1.1)^5] \\ &= -\$80{,}000 + \$18{,}181.82 + \$33{,}057.85 - \$90{,}157.77 + \$54{,}641.07 + \$62{,}092.13 \\ &= -\$2{,}184.90\end{aligned}$$

On a purely monetary basis, a single-stage expansion seems to be a better alternative than this two-stage expansion. However, other qualitative factors as mentioned earlier must be considered as well.

TABLE 6.1 | CASH FLOWS FOR TWO-STAGE EXPANSION AT GRANDMOTHER'S CHICKEN RESTAURANT

Year	Projected Demand (meals/yr)	Projected Capacity (meals/yr)	Calculation of Incremental Cash Flow Compared to Base Case (80,000 meals/yr)	Cash Inflow (outflow)
0	80,000	80,000	Increase kitchen capacity to 105,000 meals =	($80,000)
1	90,000	105,000	90,000 − 80,000 = (10,000 meals)($2/meal) =	$20,000
2	100,000	105,000	100,000 − 80,000 = (20,000 meals)($;2/meal) =	$40,000
3	110,000	105,000	105,000 − 80,000 = (25,000 meals)($2/meal) =	$50,000
			Increase total capacity to 130,000 meals =	($170,000)
				($120,000)
4	120,000	130,000	120,000 − 80,000 = (40,000 meals)($2/meal) =	$80,000
5	130,000	130,000	130,000 − 80,000 = (50,000 meals)($2/meal) =	$100,000

Discussion Questions

1. A young boy sets up a lemonade stand on the corner of College Street and Air Park Boulevard. Temperatures in the area climb to 100°F. during the summer. The intersection is near a major university and a large construction site. Explain to this young entrepreneur how his business might benefit from economies of scale. Explain also some conditions that might lead to diseconomies of scale.
2. What are the economies of scale in college class size? As class size increases, what symptoms of diseconomies of scale appear? How are these symptoms related to customer contact?

Problems

The OM Explorer and POM for Windows software is available to all students using the 10th edition of this textbook. Go to **www.pearsonglobaleditions.com/krajewski** to download these computer packages. If you purchased MyOMLab, you also have access to Active Models software and significant help in doing the following problems. Check with your instructor on how best to use these resources. In many cases, the instructor wants you to understand how to do the calculations by hand. At the least, the software provides a check on your calculations. When calculations are particularly complex and the goal is interpreting the results in making decisions, the software replaces entirely the manual calculations.

Problems 12, 16, 14, 18, 19 and 28 require reading of Supplement A, "Decision Making Models." Problems 6, 13, 17, 19 and 18 require reading of MyOMLab Supplement F, "Financial Analysis."

1. A process currently services an average of 50 customers per day. Observations in recent weeks show that its utilization is about 90 percent, allowing for just a 10 percent capacity cushion. If demand is expected to be 75 percent of the current level in 5 years and management wants to have a capacity cushion of just 5 percent, what capacity requirement should be planned?
2. The Dahlia Medical Center has 30 labor rooms, 15 combination labor and delivery rooms, 3 delivery rooms, and 1 special delivery room reserved for complicated births. All of these facilities operate around the clock. Time spent in labor rooms varies from hours to days, with an average of about a day. The average uncomplicated delivery requires about 1 hour in a delivery room.

   During an exceptionally busy 3-day period, 109 healthy babies were born at Dahlia Medical Center. Sixty babies were born in separate labor and delivery rooms, 45 were born in combined labor and delivery rooms, and only 4 babies required a labor room and the complicated-delivery room. Which of the facilities (labor rooms, combination labor and delivery rooms, or delivery rooms) had the greatest utilization rate?
3. An airline company must plan its fleet capacity and its long-term schedule of aircraft usage. For one flight

segment, the average number of customers per day is 70, which represents a 65 percent utilization rate of the equipment assigned to the flight segment. If demand is expected to increase to 84 customers for this flight segment in 3 years, what capacity requirement should be planned? Assume that management deems that a capacity cushion of 25 percent is appropriate.

4. Macon Controls produces three different types of control units used to protect industrial equipment from overheating. Each of these units must be processed by a machine that Macon considers to be their process bottleneck. The plant operates on two 8-hour shifts, 5 days per week, 52 weeks per year. Table 6.2 provides the time standards at the bottleneck, lot sizes, and demand forecasts for the three units. Because of demand uncertainties, the operations manager obtained three demand forecasts (pessimistic, expected, and optimistic). The manager believes that a 20 percent capacity cushion is best.

TABLE 6.2 | CAPACITY INFORMATION FOR MACON CONTROLS

	TIME STANDARD			DEMAND FORECAST		
Component	**Processing (hr/unit)**	**Setup (hr/lot)**	**Lot Size (units/lot)**	**Pessimistic**	**Expected**	**Optimistic**
A	0.05	1.0	60	15,000	18,000	25,000
B	0.20	4.5	80	10,000	13,000	17,000
C	0.05	8.2	120	17,000	25,000	40,000

   a. How many machines are required to meet minimum (Pessimistic) demand, expected demand, maximum (Optimistic) demand?
   b. How many machines are required if the operations manager decides to double lot sizes?
   c. If the operations manager has three machines and believes that the plant can reduce setup time by 20 percent through process improvement initiatives, does that plant have adequate capacity to meet all demand scenarios without increasing lot sizes?

5. The Astro World amusement park has the opportunity to expand its size now (the end of year 0) by purchasing adjacent property for $250,000 and adding attractions at a cost of $550,000. This expansion is expected to increase attendance by 30 percent over projected attendance without expansion. The price of admission is $30, with a $5 increase planned for the beginning of year 3. Additional operating costs are expected to be $100,000 per year. Estimated attendance for the next 5 years, *without expansion*, is as follows:

Year	1	2	3	4	5
Attendance	30,000	34,000	36,250	38,500	41,000

   a. What are the pretax combined cash flows for years 0 through 5 that are attributable to the park's expansion?
   b. Ignoring tax, depreciation, and the time value of money, determine how long it will take to recover (pay back) the investment.

6. MKM International is seeking to purchase a new CNC machine in order to reduce costs. Two alternative machines are in consideration. Machine 1 costs $400,000, but yields a 15 percent savings over the current machine used. Machine 2 costs $1,000,000, but yields a 25 percent savings over the current machine used. In order to meet demand, the following forecasted cost information for the current machine is also provided.

   a. Based on the NPV of the cash flows for these 5 years, which machine should MKM International Purchase? Assume a discount rate of 9 percent.
   b. If MKM International lowered its required discount rate to 9 percent, what machine would it purchase?

Year	**Projected Cost**
1	950,000
2	1,350,000
3	1,450,000
4	1,500,000
5	2,500,000

7. Tuff-Rider, Inc., manufactures touring bikes and mountain bikes in a variety of frame sizes, colors, and component combinations. Identical bicycles are produced in lots of 100. The projected demand, lot size, and time standards are shown in the following table:

Item	**Touring**	**Mountain**
Demand forecast	5,000 units/year	10,000 units/year
Lot size	100 units	100 units
Standard processing time	.25 hour/unit	.50 hour/unit
Standard setup time	2 hours/lot	3 hours/lot

   The shop currently works 8 hours a day, 5 days a week, 50 weeks a year. It operates five workstations, each producing one bicycle in the time shown in the table. The shop maintains a 15 percent capacity cushion. How many workstations will be required next year to meet expected demand without using overtime and without decreasing the firm's current capacity cushion?

8. Up, Up, and Away is a producer of kites and wind socks. Relevant data on a bottleneck operation in

the shop for the upcoming fiscal year are given in the following table:

Item	Kites	Wind Socks
Demand forecast	30,000 units/year	12,000 units/year
Lot size	20 units	70 units
Standard processing time	0.3 hour/unit	1.0 hour/unit
Standard setup time	3.0 hours/lot	4.0 hours/lot

The shop works two shifts per day, 8 hours per shift, 200 days per year. Currently, the company operates four machines, and desires a 25 percent capacity cushion. How many machines should be purchased to meet the upcoming year's demand without resorting to any short-term capacity solutions?

9. Kim Epson operates a full-service car wash, which operates from 8 A.M. to 8 P.M. 7 days a week. The car wash has two stations: an automatic washing and drying station and a manual interior cleaning station. The automatic washing and drying station can handle 30 cars per hour. The interior cleaning station can handle 230 cars per day. Based on a recent year-end review of operations, Kim estimates that future demand for the interior cleaning station for the 7 days of the week, expressed in average number of cars per day, would be as follows:

Day	Mon.	Tues.	Wed.	Thurs.	Fri.	Sat.	Sun.
Cars	160	180	150	140	290	270	240

By installing additional equipment (at a cost of $100,000), Kim can increase the capacity of the interior cleaning station to 300 cars per day. Each car wash generates a pretax contribution of $5.00. Should Kim install the additional equipment if she expects a pretax payback period of 3 years or less?

10. Roche Brothers is considering a capacity expansion of its supermarket. The landowner will build the addition to suit in return for $200,000 upon completion and a 5-year lease. The increase in rent for the addition is $10,000 per month. The annual sales projected through year 5 follow. The current effective capacity is equivalent to 500,000 customers per year. Assume a 2 percent pretax profit on sales.

Year	1	2	3	4	5
Customers	560,000	600,000	685,000	700,000	715,000
Average Sales per Customer	$50.00	$53.00	$56.00	$60.00	$64.00

a. If Roche expands its capacity to serve 700,000 customers per year now (end of year 0), what are the projected annual incremental pretax cash flows attributable to this expansion?

b. If Roche expands its capacity to serve 700,000 customers per year at the end of year 2, the landowner will build the same addition for $240,000 and a 3-year lease at $12,000 per month. What are the projected annual incremental pretax cash flows attributable to this expansion alternative?

11. Arabelle is considering expanding the floor area of her high-fashion import clothing store, The French Prints of Arabelle, by increasing her leased space in the upscale Cherry Creek Mall from 2,000 square feet to 3,000 square feet. The Cherry Creek Mall boasts one of the country's highest ratios of sales value per square foot. Rents (including utilities, security, and similar costs) are $110 per square foot per year. Salary increases related to French Prints' expansion are shown in the following table, along with projections of sales per square foot. The purchase cost of goods sold averages 70 percent of the sales price. Sales are seasonal, with an important peak during the year-end holiday season.

Year	Quarter	Sales (per sq ft)	Incremental Salaries
1	1	$ 90	$12,000
	2	60	8,000
	3	110	12,000
	4	240	24,000
2	1	99	12,000
	2	66	8,000
	3	121	12,000
	4	264	24,000

a. If Arabelle expands French Prints at the end of year 0, what will her quarterly pretax cash flows be through year 2?

b. Project the quarterly pretax cash flows assuming that the sales pattern (10 percent annually compounded increase) continues through year 3.

12. Dawson Electronics is a manufacturer of high-tech control modules for lawn sprinkler systems. Denise, the CEO, is trying to decide if the company should develop one of the two potential new products, the Water Saver 1000 or the Greener Grass 5000. With each product, Dawson can capture a bigger market share if it chooses to expand capacity by buying additional machines. Given different demand scenarios, their probabilities of occurrence, and capacity expansion versus no change in capacity, the potential sales of each product are summarized in Table 6.3.

TABLE 6.3 DEMAND AND SALES INFORMATION FOR DAWSON ELECTRONICS

	Water Saver 1000 Dollar Sales ($1000)	Greener Grass 5000 Dollar Sales ($1000)	Probability of Occurrence
With Capacity Expansion			
Low Demand	1000	2500	0.25
Medium Demand	2000	3000	0.50
High Demand	3000	5000	0.25
Without Capacity Expansion			
Low Demand	700	1000	0.25
Medium Demand	1000	2000	0.50
High Demand	2000	3000	0.25

a. What is the expected payoff for Water Saver 1000 and the Greener Grass 5000, with and without capacity expansion?

b. Which product should Denise choose to produce, and with which capacity expansion option?

Advanced Problems

13. Several years ago, River City built a water purification plant to remove toxins and filter the city's drinking water. Because of population growth, the demand for water next year will be more than the plant's capacity of 120 million gallons per year. Therefore, the city must expand the facility. The estimated demand over the next 20 years is given in Table 6.4.

The city planning commission is considering three alternatives.

- *Alternative 1:* Expand enough at the end of year 0 to last 20 years, which means an 80 million gallon increase (200 –120).
- *Alternative 2:* Expand at the end of year 0 and at the end of year 10.
- *Alternative 3:* Expand at the end of years 0, 5, 10, and 15.

Each alternative would provide the needed 200 million gallons per year at the end of 20 years, when the value of the plant would be the same regardless of the alternative chosen. Significant economies of scale can be achieved in construction costs: A 20 million gallon expansion would cost $18 million; a 40 million gallon expansion, $30 million; and an 80 million gallon expansion, only $50 million. The level of future interest rates is uncertain, leading to uncertainty about the hurdle rate. The city believes that it could be as low as 12 percent and as high as 16 percent (see MyOMLab Supplement F, "Financial Analysis").

a. Compute the cash flows for each alternative, compared to a base case of doing nothing. (*Note:* As a municipal utility, the operation pays no taxes.)

b. Which alternative minimizes the present value of construction costs over the next 20 years if the discount rate is 12 percent? Sixteen percent?

c. Because the decision involves public policy and compromise, what political considerations does the planning commission face?

TABLE 6.4 | WATER DEMAND

Year	Demand	Year	Demand	Year	Demand
0	120	7	148	14	176
1	124	8	152	15	180
2	128	9	156	16	184
3	132	10	160	17	188
4	136	11	164	18	192
5	140	12	168	19	196
6	144	13	172	20	200

14. Acme Steel Fabricators experienced booming business for the past 5 years. The company fabricates a wide range of steel products, such as railings, ladders, and light structural steel framing. The current manual method of materials handling is causing excessive inventories and congestion. Acme is considering the purchase of an overhead rail-mounted hoist system or a forklift truck to increase capacity and improve manufacturing efficiency.

The annual pretax payoff from the system depends on future demand. If demand stays at the current level, the probability of which is 0.70, annual savings from the overhead hoist will be $10,000. If demand rises, the hoist will save $22,000 annually because of operating efficiencies in addition to new sales. Finally, if demand falls, the hoist will result in an estimated annual loss of $50,000. The probability is estimated to be 0.20 for higher demand and 0.10 for lower demand.

If the forklift is purchased, annual payoffs will be $5,000 if demand is unchanged, $15,000 if demand rises, and −$20,000 if demand falls.

a. Draw a decision tree for this problem and compute the expected value of the payoff for each alternative.

b. Which is the best alternative, based on the expected values?

15. Knott's Industries manufactures standard and super premium backyard swing sets. Currently it has four identical swing-set-making machines, which are operated 250 days per year and 8 hours each day. A capacity cushion of 20 percent is desired. The following information is also known:

	Standard Model	Super Premium Model
Annual Demand	20,000	10,000
Standard Processing Time	7 min	20 min
Average Lot Size	50	30
Standard Setup Time per Lot	30 min	45 min

a. Does Knott's have sufficient capacity to meet annual demand?

b. If Knott's was able to reduce the setup time for the Super Premium Model from 45 minutes to 30 minutes, would there be enough current capacity to produce 20,000 units of each type of swing set?

16. A manager is trying to decide whether to buy one machine or two. If only one machine is purchased and demand proves to be excessive, the second machine can be purchased later. Some sales would be lost, however, because the lead time for delivery of this type of machine is 6 months. In addition, the cost per machine will be lower if both machines are purchased at the same time. The probability of low demand is estimated to be 0.20 and that of high demand to be 0.80. The after-tax NPV of the benefits from purchasing two machines together is $70,000 if demand is low and $160,000 if demand is high.

If one machine is purchased and demand is low, the NPV is $120,000. If demand is high, the manager has three options: (1) doing nothing, which has an NPV of $100,000; (2) subcontracting, with an NPV of $150,000; and (3) buying the second machine, with an NPV of $140,000.

a. Draw a decision tree for this problem.

b. What is the best decision and what is its expected payoff?

17. Mars Incorporated is interested in going to market with a new fuel savings device that attaches to electrically powered industrial vehicles. The device, code named

"Python," promises to save up to 15 percent of the electrical power required to operate the average electric forklift. Mars expects that modest demand expected during the introductory year will be followed by a steady increase in demand in subsequent years. The extent of this increase in demand will be based on customer's expectations regarding the future cost of electricity, and which is shown in Table 6.5. Mars expects to sell the device for $500 each, and does not expect to be able to raise its price over the foreseeable future.

TABLE 6.5 DEMAND FOR PYTHON POWER SAVING DEVICE

	EXPECTED DEMAND OF THE DEVICE IN UNITS/YEAR	
Year	Small Increases in the Cost of Electrical Power	Large Increases in the Cost of Electrical Power
1	1,000	10,000
2	5,000	8,000
3	1,000	15,000
4	15,000	20,000
5	18,000	30,000

Mars is faced with two alternatives:

- *Alternative 1:* Make the device themselves, which requires an initial outlay of $250,000 in plant and equipment and a variable cost of $75 per unit.
- *Alternative 2:* Outsource the production, which requires no initial investment, but incurs a per unit cost of $300.

a. Assuming small increases in the cost of electrical power, compute the cash flows for each alternative. Over the next 5 years, which alternative maximizes the NPV of this project if the discount rate is 10 percent?

b. Assuming large increases in the cost of electrical power, compute the cash flows for each alternative. Over the next 5 years, which alternative maximizes the NPV of this project if the discount rate is 10 percent?

18. The vice president of operations at Dintell Corporation, a major supplier of passenger-side automotive air bags, is considering a $50 million expansion at the firm's Fort Worth, Texas, production complex. The most recent economic projections indicate a 0.60 probability that the overall market will be $400 million per year over the next 5 years and a 0.40 probability that the market will be only $200 million per year during the same period. The marketing department estimates that Dintell has a 0.50 probability of capturing 40 percent of the market and an equal probability of obtaining only 30 percent of the market. The cost of goods sold is estimated to be 70 percent of sales. For planning purposes, the company currently uses a 12 percent discount rate, a 40 percent tax rate, and the MACRS depreciation schedule. The criteria for investment decisions at Dintell are (1) the net expected present value must be greater than zero; (2) there must be at least a 70 percent chance that the net present value will be positive; and (3) there must be no more than a 10 percent chance that the firm will lose more than 20 percent of the initial value.

a. Based on the stated criteria, determine whether Dintell should fund the project.

b. What effect will a probability of 0.70 of capturing 40 percent of the market have on the decision?

c. What effect will an increase in the discount rate to 15 percent have on the decision? A decrease to 10 percent?

d. What effect will the need for another $10 million in the third year have on the decision?

19. Darren Mack owns the "Gas n' Go" convenience store and gas station. After hearing a marketing lecture, he realizes that it might be possible to draw more customers to his high-margin convenience store by selling his gasoline at a lower price. However, the "Gas n' Go" is unable to qualify for volume discounts on its gasoline purchases, and therefore cannot sell gasoline for profit if the price is lowered. Each new pump will cost $95,000 to install, but will increase customer traffic in the store by 1,000 customers per year. Also, because the "Gas n' Go" would be selling its gasoline at no profit, Darren plans on increasing the profit margin on convenience store items incrementally over the next 5 years. Assume a discount rate of 8 percent. The projected convenience store sales per customer and the projected profit margin for the next 5 years are as follows:

Year	Projected Convenience Store Sales Per Customer	Projected Profit Margin
1	$ 5.00	20%
2	$ 6.50	25%
3	$ 8.00	30%
4	$10.00	35%
5	$11.00	40%

a. What is the NPV of the next 5 years of cash flows if Darren had four new pumps installed?

b. If Darren required a payback period of 4 years, should he go ahead with the installation of the new pumps?

20. Mackelprang, Inc., is in the initial stages of building the premier planned community in the greater Phoenix, Arizona, metropolitan area. The main selling point will be the community's lush golf courses. Homes with golf course views will generate premiums far larger than homes with no golf course views, but building golf courses is expensive and takes up valuable space that non-view homes could be built upon. Mackelprang, Inc., has limited land capacity. In order to maximize its profits, it is faced with a decision as to how many golf courses it should build, which, in turn, will impact how many homes with and without golf course views it will be able to construct. Mackelprang, Inc., realizes that this decision is directly related to the premium buyers will be willing to spend to buy homes with golf course views. Mackelprang, Inc., is required to build at least one golf course, but has enough space to build up to three golf courses. The following table indicates the costs and potential revenues for each course:

a. Which golf course or courses should Mackelprang, Inc., build?

b. What is the expected payoff for this project?

	Indian River	The Cactus	Wildwood
Cost	$2.2M	$2.5M	$3M
Highest Possible Revenue	$4M	$4M	$5M
Probability of High Revenue	0.2	0.3	0.3
Likely Revenue	$1.5M	$3M	$3M
Probability of Likely Revenue	0.5	0.4	0.5
Lowest Possible Revenue	$1M	$1M	$1M
Probability of Low Revenue	0.3	0.3	0.2

21. Two new alternatives have come up for expanding Grandmother's Chicken Restaurant (see Solved Problem 2). They involve more automation in the kitchen and feature a special cooking process that retains the original-recipe taste of the chicken. Although the process is more capital-intensive, it would drive down labor costs, so the pretax profit for *all* sales (not just the sales from the capacity added) would go up from 20 to 22 percent. This gain would increase the pretax profit by 2 percent of each sales dollar through $800,000 (80,000 meals × $10) and by 22 percent of each sales dollar between $800,000 and the new capacity limit. Otherwise, the new alternatives are much the same as those in Example 6.2 and Solved Problem 2.

- *Alternative 1:* Expand both the kitchen and the dining area now (at the end of year 0), raising the capacity to 130,000 meals per year. The cost of construction, including the new automation, would be $336,000 (rather than the earlier $200,000).
- *Alternative 2:* Expand only the kitchen now, raising its capacity to 105,000 meals per year. At the end of year 3, expand both the kitchen and the dining area to the 130,000 meals-per-year volume. Construction and equipment costs would be $424,000, with $220,000 at the end of year 0 and the remainder at the end of year 3. As with alternative 1, the contribution margin would go up to 22 percent.

With both new alternatives, the salvage value would be negligible. Compare the cash flows of all alternatives. Should Grandmother's Chicken Restaurant expand with the new or the old technology? Should it expand now or later?

22. Referring to problem 4, the operations manager at Macon Controls believes that pessimistic demand has a probability of 20 percent, expected demand has a probability of 50 percent, and optimistic demand has a probability of 30 percent. Currently, new machines must be purchased at a cost of $500,000 a piece, the price charged for each control unit is $110, and the variable cost of production is $50 per unit. (Hint: since the price and variable cost for each control unit are the same, the profit maximizing product mix will be the same as the mix that maximizes the total number of units produced.)

a. Draw a decision tree for this problem.

b. How many machines should the company purchase, and what is the expected payoff?

VIDEO CASE Gate Turnaround at Southwest Airlines

Rollin King and Herb Kelleher started Southwest Airlines in 1971 with this idea: if they could take airline passengers where they want to go, on time, at the lowest possible price, and have a good time while doing it, people would love to fly their airline. The result? No other airline in the industry's history has enjoyed the customer loyalty and extended profitability for which Southwest is now famous. The company now flies more than 3,400 times each day to over 64 destinations across the United States.

There's more to the story, however, than making promises and hoping to fulfill them. A large part of Southwest Airlines' success lies in its ability to plan long-term capacity to better match demand, as also improving the utilization of its fleet by turning around an aircraft at the gate faster than its competitors. Capacity at Southwest is measured in seat-miles, and even a single minute reduction in aircraft turnaround time system wide means additional seat-miles being added to the available capacity of Southwest Airlines.

As soon as an aircraft calls "in range" at one of Southwest's airport locations, called a station, the local operations manager notifies the ground operations team so that they can start mobilizing all the parties involved in servicing the aircraft in preparation for its next departure. The grounds operations team consists of a baggage transfer driver who has responsibility for getting connecting flight bags to their proper planes, a local baggage driver who moves bags to baggage claim for passenger pick-up, a lavatory truck driver who handles restroom receptacle drainage, a lead gate agent to handle baggage carts and track incoming and outgoing bag counts, and a bin agent to manage baggage and cargo inside the plane. The ground operations team knows it must turn the plane around in 25 minutes or less. The clock starts when the pilot sets the wheel brakes.



Pearson

Baggage transfer starts less than 40 seconds after engine shutdown at Southwest Airlines.

Inbound and outbound flights are coordinated by the supervisors between Southwest's 64 airport stations through the company's Operations Terminal Information System (OTIS). Each local supervisor is able to keep track of their flights and manage any delays or problems that may have crept into the system by keeping in touch with headquarters in Dallas for system-wide issues that may impact a local station, along with using the OTIS information coming from stations sending flights their way.

Just what, exactly, does it take to turn around an aircraft? In-bound flight 3155 from Phoenix to Dallas' Love Field is a good example. In Phoenix, the operations coordinators and ground operations team push back the plane as scheduled at 9:50 A.M. The flight is scheduled to arrive at 3:35 P.M. in Dallas. The Phoenix team enters into OTIS the information the ground operations team will need in Dallas, such as wheelchairs, gate-checked baggage, cargo bin locator data, and other data needed to close out the flight on their end. This action lets the Dallas station know what to expect when the plane lands.

In Dallas, the local ground operations coordinators have been monitoring all 110 inbound flights and now see Phoenix flight 3155 in the system, scheduled for an on-time arrival. When the pilot calls "in-range" as it nears Dallas, the ground crew prepares for action.

As the plane is guided to its "stop mark" at the gate, the lead agent waits for the captain's signal that the engines have been turned off and brakes set. Within just 10 seconds, the provisioning truck pulls up to open the back door for restocking supplies such as drinks and snacks. The waiting fuel truck extends its hose to the underwing connection, and in less than 2 minutes, picks up refueling instructions and starts to load fuel. As soon as the aircraft is in position, the operations team steers the jetway into position and locks it against the aircraft. The door is opened, the in-flight crew is greeted, and passengers start to deplane.

Outside, less than 40 seconds after engine shutdown, baggage is rolling off the plane and gets placed onto the first cart. Any transfer bags get sent to their next destination, and gate-checked bags are delivered to the top of the jetway stairs for passenger pick-up.

While passengers make their way out of the plane, the in-flight crew helps clean up and prepare the cabin for the next flight. If all goes well, the last passenger will leave the plane after only 8 minutes. By this time, passengers waiting to board have already lined up in their designated positions for boarding. The gate agent confirms that the plane is ready for passenger boarding, and calls for the first group to turn in their boarding passes and file down the jetway.

At the completion of boarding, the operations agent checks the fuel invoice, cargo bin loading schedule with actual bag counts in their bins from the baggage agents, and a lavatory service record confirming that cleaning has taken place. Final paperwork is given to the captain. The door to the aircraft is closed, and the jetway is retracted. Thirty seconds later, the plane is pushed back and the operations agent gives a traditional salute to the captain to send the flight on its way. Total elapsed time: less than 25 minutes.

Managing Southwest's capacity has been somewhat simplified by strategic decisions made early on in the company's life. First, the company's fleet of aircraft is all Boeing 737's. This single decision impacts all areas of operations — from crew training to aircraft maintenance. The single-plane configuration also provides Southwest with crew scheduling flexibility. Since pilots and flight crews can be deployed across the entire fleet, there are no constraints with regard to training and certification pegged to specific aircraft types.

The way Southwest has streamlined its operations for tight turnarounds means it must maintain a high capacity cushion to accommodate variability in its daily operations. Anything from weather delays to unexpected maintenance issues at the gate can slow down the flow of operations to a crawl. To handle these unplanned but anticipated challenges, Southwest builds into its schedules enough cushion to manage these delays yet not so much that employees and planes are idle. Additionally, the company encourages discussion to keep on top of what's working and where improvements can be made. If a problem is noted at a downstream station, say bags were not properly loaded, this information quickly travels back up to the originating station for correction so that it does not happen again.

Even with the tightly managed operations Southwest Airlines enjoys, company executives know that continued improvement is necessary if the company is to remain profitable into the future. Company executives know when they have achieved their goals when internal and external metrics are reached. For example, the Department of Transportation (DOT) tracks on-time departures, customer complaints, and mishandled baggage for all airlines. The company sets targets for achievement on these dimensions and lets employees know on a monthly basis how the company is doing against those metrics and the rest of the industry. Regular communication with all employees is delivered via meetings, posters, and newsletters. Rewards such as prizes and profit sharing are given for successful achievement.

As for the future, Bob Jordan, Southwest's Executive Vice President for Strategy and Planning, puts it this way: "We make money when our planes are in the air, not on the ground. If we can save one minute off every turn system-wide, that's like putting five additional planes in the air. If a single plane generates annual revenue of $25 million, there's $125 million in profit potential from those time savings."

QUESTIONS

1. How can capacity and utilization be measured at an airline such as Southwest Airlines?
2. Which factors can adversely impact turn-around times at Southwest Airlines?
3. How does Southwest Airlines know they are achieving their goals?
4. What are the important long-term issues relevant for managing capacity, revenue, and customer satisfaction for Southwest Airlines?

CASE Fitness Plus, Part A

Fitness Plus, Part B, explores alternatives to expanding a new downtown facility and is included in the Instructor's Resource Manual. If you are interested in this topic, ask your instructor for a preview.

Fitness Plus is a full-service health and sports club in Greensboro, North Carolina. The club provides a range of facilities and services to support three primary activities: fitness, recreation, and relaxation. Fitness activities generally take place in four areas of the club: the (1) aerobics room, which can accommodate 35 people per class; a (2) room equipped with free weights; a (3) workout room with 24 pieces of Nautilus equipment; and a (4) large workout room containing 29 pieces of cardiovascular equipment. This equipment includes nine stairsteppers, six treadmills, six life-cycle bikes, three airdyne bikes, two cross-aerobics machines, two rowing machines, and one climber. Recreational facilities comprise eight racquetball courts, six tennis courts, and a large outdoor pool. Fitness Plus also sponsors softball, volleyball, and swim teams in city recreation leagues. Relaxation is accomplished through yoga classes held twice a week in the aerobics room, whirlpool tubs located in each locker room, and a trained massage therapist.

Situated in a large suburban office park, Fitness Plus opened its doors in 1995. During the first 2 years, membership was small and use of the facilities was light. By 1997, membership had grown as fitness began to play a large role in more and more people's lives. Along with this growth came increased use of club facilities. Records indicate that in 2000, an average of 15 members per hour checked into the club during a typical day. Of course, the actual number of members per hour varied by both day and time. On some days during a slow period, only six to eight members would check in per hour. At a peak time, such as Mondays from 4:00 P.M. to 7:00 P.M., the number would be as high as 40 per hour.

The club was open from 6:30 A.M. to 11:00 P.M. Monday through Thursday. On Friday and Saturday, the club closed at 8:00 P.M., and on Sunday the hours were 12:00 P.M. to 8:00 P.M.

As the popularity of health and fitness continued to grow, so did Fitness Plus. By May 2005, the average number of members arriving per hour during a typical day had increased to 25. The lowest period had a rate of 10 members per hour; during peak periods, 80 members per hour checked in to use the facilities. This growth brought complaints from members about overcrowding and unavailability of equipment. Most of these complaints centered on the Nautilus, cardiovascular, and aerobics fitness areas. The owners began to wonder whether the club was indeed too small for its membership. Past research indicated that individuals work out an average of 60 minutes per visit. Data collected from member surveys showed the following facilities usage pattern: 30 percent of the members do aerobics, 40 percent use the cardiovascular equipment, 25 percent use the Nautilus machines, 20 percent use the free weights, 15 percent use the racquetball courts, and 10 percent use the tennis courts. The owners wondered whether they could use this information to estimate how well existing capacity was being utilized.

If capacity levels were being stretched, now was the time to decide what to do. It was already May, and any expansion of the existing facility would take at least 4 months. The owners knew that January was always a peak membership enrollment month and that any new capacity needed to be ready by then. However, other factors had to be considered. The area was growing both in terms of population and geographically. The downtown area just received a major facelift, and many new offices and businesses were moving back to it, causing a resurgence in activity.

With this growth came increased competition. A new YMCA was offering a full range of services at a low cost. Two new health and fitness facilities had opened within the past year in locations 10 to 15 minutes from Fitness Plus. The first, called the Oasis, catered to the young adult crowd and restricted the access of children under 16 years old. The other facility, Gold's Gym, provided excellent weight and cardiovascular training only.

As the owners thought about the situation, they had many questions: Were the capacities of the existing facilities constrained, and if so, where? If capacity expansion was necessary, should the existing facility be expanded? Because of the limited amount of land at the current site, expansion of some services might require reducing the capacity of others. Finally, owing to increased competition and growth downtown, was now the time to open a facility to serve that market? A new facility would take 6 months to renovate, and the financial resources were not available to do both.

QUESTIONS

1. What method would you use to measure the capacity of Fitness Plus? Has Fitness Plus reached its capacity?
2. Which capacity strategy would be appropriate for Fitness Plus? Justify your answer.
3. How would you link the capacity decision being made by Fitness Plus to other types of operating decisions?

Selected References

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